

GENERAL SCIENCE GRADE 10: MICROORGANISMS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.7 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 1.0: Life Science
Sub-Strand	Sub-Strand 1.7: Microorganisms
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Types of microorganisms that affect human beings (fungi, bacteria and viruses) – Mode of transmission of microorganisms in human beings – Infections caused by microorganisms in human beings (fungal, bacterial and viral infections) – Methods of control and prevention of infections caused by microorganisms – Economic importance of microorganisms – Growth of microorganisms on food substrates (mould formation) – Food preservation and its significance to the community
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify microorganisms that affect human beings, - explain modes of transmission of microorganisms in human beings, - identify types of infections caused by microorganisms in human beings, - describe methods of control and prevention of infections caused by microorganisms, - explain the economic importance of microorganisms, - appreciate the effect of microorganisms in day-to-day life.
Core Competencies	– Digital literacy: as the learner uses digital devices to research on infections caused by microorganisms as well as their control and prevention mechanisms. – Communication and collaboration: as the learner shares information gathered on types of microorganism infections, control and prevention with peers in plenary discussions and group investigations. – Critical thinking and problem solving: as the learner analyses how the mode of transmission of a microorganism determines the most effective control and prevention strategy. – Self-efficacy: as the learner designs and carries out a project to investigate the growth of microorganisms (mould formation) on food substrates such as ugali, bread and maize flour.
Core Values	– Respect: as the learner becomes considerate of others' opinions while discussing methods of controlling and preventing microorganism infections in group settings. – Responsibility: as the learner handles food substrates and biological samples safely during the mould-growth investigation, disposing of materials appropriately to prevent environmental contamination.

	– Integrity: as the learner records and reports observations honestly during the investigation of microorganism growth on food substrates. – Unity: as the learner collaborates equitably with peers during group investigations and community sensitisation activities.
Science & Engineering Practices	– Asking Questions and Defining Problems: as the learner formulates investigable questions about why certain foods (e.g., ugali left overnight) spoil faster than others and what microorganisms are responsible. – Planning and Carrying Out Investigations: as the learner designs and conducts an experiment to observe mould growth on different food substrates under varied conditions of moisture and temperature. – Analysing and Interpreting Data: as the learner records, compares and draws conclusions from observations of microbial growth rates on different food substrates. – Constructing Explanations: as the learner uses evidence from investigations and research to explain how transmission routes determine control methods for fungal, bacterial and viral infections. – Obtaining, Evaluating and Communicating Information: as the learner uses digital and print media to research infections and then communicates findings to peers and the community on food preservation and microorganism significance.
Pertinent & Contemporary Issues (PCIs)	– Socio-economic issues (poverty eradication): as the learner sensitises the community on the significance of food preservation and the economic importance of microorganisms, linking microbial activity to livelihoods of Kenyan farmers and traders in markets such as Wakulima Market, Nairobi. – Health education: as the learner discusses how modes of transmission of bacteria, viruses and fungi drive prevention measures relevant to everyday Kenyan household and school settings. – Environmental conservation: as the learner responsibly manages biological waste from the mould-growth investigation to avoid pollution of the school compound and surrounding community.
Career Connections	– Medical Laboratory Technologist: this sub-strand introduces identification and culturing of microorganisms, foundational to diagnostic laboratory work in Kenyan health facilities. – Public Health Officer: understanding modes of transmission and prevention of microbial infections underpins community health campaigns and disease surveillance roles in Kenya. – Food Scientist / Food Technologist: knowledge of microbial spoilage and food preservation connects directly to careers in Kenya's food processing and value-addition industry (e.g., Brookside Dairy, Unga Group). – Microbiologist / Research Scientist: the investigation into mould growth and economic importance of microorganisms provides a foundation for academic and applied microbiology research. – Agronomist / Agricultural Extension Officer: understanding beneficial microorganisms (e.g., nitrogen-fixing bacteria, fungi in composting) is essential for advising farmers in the Rift Valley and Central Kenya on soil health.
Focus for Lessons	This unit investigates the world of microorganisms — fungi, bacteria and viruses — through the lens of their interaction with human life in Kenya. Learners move from identifying microbial types and their transmission routes, to exploring infections, prevention strategies and the surprising economic value of microorganisms in food production, medicine and agriculture. A project-based investigation of mould growth on local food substrates (ugali, chapati, sukuma wiki) anchors the learning in observable, everyday phenomena, while a community sensitisation task on food preservation connects classroom science to household and economic relevance.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why did the ugali and chapati grow fuzzy patches overnight — and how can understanding the invisible organisms responsible help us protect our health and our food in Kenya? KICD KEY INQUIRY QUESTION: 1. How are microorganisms important in day-to-day life?

Anchoring Phenomenon	A Standard stainless-steel sufuria of leftover ugali and a half-eaten chapati are stored uncovered on a kitchen counter in Kisumu overnight. By morning, the family notices white, green and black fuzzy patches growing on the food, an unpleasant sour smell, and some areas turning slimy. The ugali has not been touched by anyone, no insects are visible, and the kitchen appeared clean. Students are shown photographs of the spoiled food and asked: where did these patches come from, why did they appear overnight, are they alive, and are all of them the same kind of organism? The phenomenon is puzzling because the growth seems to arise from nothing, it happened fast, it varies in colour and texture across the same food surface, and students must grapple with the invisible world of microorganisms to explain what they are observing.
Storyline Thread	Lesson 1 — Noticing and Wondering (Predict Phase): Students are shown photographs of the spoiled ugali and chapati phenomenon. In pairs they write predictions about what the patches are, where they came from and whether they are alive. Class shares predictions; teacher records them on the Driving Question Board (DQB). Students brainstorm prior knowledge of "tiny organisms" and the class maps out what they already know and what they need to find out. Lesson 2 — Types of Microorganisms (Observe Phase 1): Guided research using digital devices and/or print media to identify the three types of microorganisms affecting human beings — fungi, bacteria and viruses. Students observe photomicrographs and illustrations, classify the fuzzy patches from the phenomenon as fungal moulds, and create a comparison table of structural features. Teacher introduces size scale using Kenyan context analogies (e.g., if a maize grain were a bacterium, a football pitch represents a human cell). Lesson 3 — Modes of Transmission (Observe Phase 2): Students use digital or print media to research modes of transmission of microorganisms (direct contact, droplets, contaminated water/food, vectors, airborne). Groups are assigned one transmission route each, research it and present findings. Class maps how the mould spores in the phenomenon were transmitted through the air onto the uncovered ugali, connecting back to the anchoring phenomenon on the DQB. Lesson 4 — Infections Caused by Microorganisms (Explain Phase 1): Structured discussion and jigsaw activity on fungal infections (e.g., ringworm, oral thrush), bacterial infections (e.g., cholera, typhoid, tuberculosis — referencing Lake Victoria cholera context) and viral infections (e.g., influenza, HIV, COVID-19). Students link each infection to its mode of transmission identified in Lesson 3. DQB updated: students revise earlier predictions and add new sub-questions. Lesson 5 — Prevention and Control (Explain Phase 2): Students discuss how mode of transmission determines control measures. Groups design a prevention poster for one infection category relevant to their local Kenyan community. Teacher facilitates analysis of the ugali phenomenon: covering food, refrigeration, and hygiene as practical prevention strategies. Students evaluate which control methods are accessible to low-income households in rural and urban Kenya. Lesson 6 — Economic Importance of Microorganisms (Explain Phase 3): Class discussion and peer-sharing on the beneficial roles of microorganisms — yoghurt and fermented uji production (<i>Lactobacillus</i>), bread and mandazi leavening (yeast/<i>Saccharomyces</i>), biogas generation, nitrogen fixation for Kenyan farms, antibiotic production, and decomposition for soil fertility. Students contrast this with harmful economic impacts (crop diseases, food spoilage losses for market vendors). DQB updated to reflect the dual nature of microorganisms. Lesson 7 — Project Investigation: Mould Growth on Food Substrates (Observe + Analyse Phase): Students set up an investigation placing small samples of ugali, bread, and a dry maize biscuit in labelled containers under different conditions (covered/uncovered, moist/dry, warm/cool). Variables are identified and controlled. Students record initial observations and predictions. Where real-time growth is not yet visible, teacher provides pre-prepared samples incubated over 2–3 days. Students observe, photograph, and record growth differences; analyse data to draw conclusions about conditions favouring microbial growth and connect to food

	preservation principles. Lesson 8 — Model Building, DQB Synthesis and Community Sensitisation (Model Building + DQB Closure): Students revise their initial models of the phenomenon (drawn in Lesson 1) to include: microbial types, spore dispersal as transmission route, conditions for growth, infection pathways, and control/prevention strategies. Groups present their final models. Class revisits the full DQB, confirms or revises all questions and predictions. Students draft a short community sensitisation message on food preservation for households in their locality, completing the KICD project requirement. Lesson closes with reflection on how the driving question has been answered.
Supporting Phenomena	– A child in a Nairobi primary school develops white patches inside the mouth (oral thrush) after a course of antibiotics — classmates observe photographs and ask why medicine itself seemed to cause a new infection, linking to fungal opportunistic infections. – A Mombasa fishing community reports that fish left unrefrigerated for a few hours smell strongly of ammonia and become unsafe to eat far faster than fish stored on ice — linking bacterial decomposition rates to temperature and food safety. – During the 2023 cholera outbreak along the shores of Lake Victoria, communities using untreated water fell ill while those boiling water did not — providing a real Kenyan context for bacterial transmission via contaminated water and prevention through simple heat treatment. – Kenyan farmers in Kericho use yoghurt starter cultures and observe that fresh milk transforms into a thick, tangy product overnight — demonstrating the beneficial economic role of bacteria (<i>Lactobacillus</i>) and connecting to the dairy value chain.

LESSON 1 (40 minutes): Noticing and Wondering: What Is Growing on Our Ugali and Chapati?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon — spoiled ugali and chapati in a Kisumu kitchen — by eliciting students' prior knowledge, generating predictions, and opening the Driving Question Board so that every subsequent lesson in the sub-strand is framed as an answer to questions students themselves have posed.
Knowledge	Students will be able to identify that microorganisms (fungi, bacteria and viruses) exist that affect human beings, and that visible signs of food spoilage (fuzzy patches, slime, sour smell) are caused by living organisms too small to see with the naked eye.
Skills	Students will practise the NGSS Science and Engineering Practice of Asking Questions and Defining Problems by generating testable questions from the phenomenon, and Obtaining, Evaluating and Communicating Information by sharing and recording prior knowledge on the Driving Question Board.
Attitudes	Students will appreciate that invisible organisms have real, observable effects on day-to-day life in Kenya, and will show respect for peers' ideas when discussing predictions.
Key Inquiry Question	How are microorganisms important in day-to-day life?
Purpose in Storyline	Lesson 1 opens the entire sub-strand storyline. Students encounter the anchoring phenomenon (spoiled ugali and chapati) for the first time, make un-coached predictions about the fuzzy patches, and collectively build the first version of the Driving Question Board. Every future lesson will return to this board to mark growing understanding. No formal teaching of content occurs yet — the lesson is entirely devoted to noticing, wondering, and surfacing prior knowledge so the teacher can plan subsequent lessons responsively.
Safety Notes	No physical food samples are handled in Lesson 1 — only photographs are used. Remind students never to smell, taste or handle mouldy food at home. If any student mentions a family member being ill from food poisoning, acknowledge sensitively and note it as motivation for the sub-strand.

B. LESSON OVERVIEW

Students are shown two high-resolution photographs of a sufuria of leftover ugali and a half-eaten chapati stored overnight on a kitchen counter in Kisumu. Both foods display white, green and black fuzzy patches, slimy areas and are described as having a sour smell. Working in pairs, students write un-coached predictions about what the patches are, where they came from, whether they are alive, and whether all patches are the same. Predictions are shared in a whole-class gallery and recorded on the newly opened Driving Question Board (DQB). The class then maps collective prior knowledge using a Know / Wonder / Notice (KWN) chart, and the lesson closes with students voting on the three most puzzling questions to anchor the sub-strand investigation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Populations.	Students enter the room to find two large printed photographs placed on their desks — one of a	As students write silently, circulate and READ (do not comment on) responses — note 3–4 interesting	SILENT THINK → PAIR → COLD-CALL SHARE (Think-Pair-Share variant). This strategy surfaces the	Observable indicator: Every student has written at least three sentences in their exercise book

communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: The FDA and Food Safety (PLIX) Source: CK-12 Search ARES for similar readings	sufuria of leftover ugali and one of a half-eaten chapati, both showing colourful fuzzy patches and slimy areas. Working individually for 2 minutes (silent think time), each student writes three quick-fire responses in their exercise books: (1) What do you think the patches are? (2) Where did they come from — no one touched the food? (3) Are the patches alive? Students then turn to their seat partner and compare their three answers for 2 minutes, noting where they agree and where they differ. Pairs prepare one shared 'best prediction' statement to report to the class.	or contrasting predictions to cold-call later. After 2 minutes say: 'Now turn to your partner and share your three ideas. You have 2 minutes — go.' [Allow 2 minutes.] Bring the class back: 'I'm going to cold-call five pairs — not to judge your answer but to collect our class's range of ideas.' Cold-call 5 pairs using a lolly-stick draw. After each response, record the key phrase on the whiteboard under the heading PREDICTIONS without comment or correction. Use probing stems: 'Can you say more about where you think it came from?' and 'What makes you think it might — or might not — be alive?' Allow 10 seconds of WAIT TIME after each probe before accepting a response. Do not confirm or deny any prediction — say: 'That is a fascinating idea — we will find out together.'	full range of prior conceptions — including misconceptions such as 'the food rotted on its own' or 'it came from the air being dirty' — before any instruction occurs. Recording all predictions without judgment creates psychological safety and gives the teacher a formative baseline of where students' mental models currently sit.	before pair-share begins. Teacher scans books during circulation — look specifically for (a) students who reference 'germs' or 'bacteria' by name (prior knowledge present), (b) students who attribute the patches to chemical reactions or dirt (misconception to address in Lesson 3), and (c) students who explicitly wonder whether different patch colours mean different organisms (readiness for classification in Lesson 2). Mentally flag at least one student from each category for targeted questions in later phases.
Observe Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: The FDA and Food Safety (PLIX) Source: CK-12 Search ARES for similar readings	The teacher distributes an 'Observation Record' half-sheet (or students use a ruled section of their exercise book). Students are directed to study both photographs again, this time using a structured NOTICE / INFER / WONDER protocol: (a) NOTICE — list at least four specific, observable things you can see in the photograph (colour, texture, location on the food, pattern of spread); (b) INFER — for each observation, write one possible explanation; (c) WONDER — write one question that your observation raises. Students work individually for 4 minutes, then share 'noticings' in a round-robin with their pair partner (1 minute each direction). The class then does a 3-minute whole-group 'Phenomena Gallery' where the teacher points	Project the two photographs on the board (or display printed A3 copies). Say: 'We are going to be scientists right now. Scientists do not jump to explanations — they first describe carefully what they observe. Look at this photograph. Tell me only what you can SEE — not what you think it is.' Point to the green patch on the ugali: 'What do you notice about the colour here? Is it uniform or patchy?' [WAIT 10 seconds.] Cold-call 3 students. Then point to the black patch on the chapati edge: 'Is this patch the same texture as the green one — describe it.' [WAIT 10 seconds.] Cold-call 2 students. Then ask: 'Some of you wrote that these patches came from the air. What specific evidence in this photograph supports — or challenges — that idea?' [WAIT 15	NOTICE / INFER / WONDER (NIW) Protocol. NIW trains students to separate empirical observation from interpretation — a foundational scientific practice (NGSS SEP 1: Asking Questions; SEP 3: Planning and Carrying Out Investigations). By forcing students to articulate what they literally see before explaining it, the teacher interrupts the common habit of jumping to conclusions and models the discipline of evidence-based reasoning that will underpin every investigation in this sub-strand.	Observable indicator: During the Phenomena Gallery, students describe physical properties (colour, texture, location, pattern) rather than causes. If students cannot separate observation from inference (e.g., they say 'I see bacteria' rather than 'I see a fuzzy green patch'), the teacher gently redirects: 'That is an inference — what do you actually see with your eyes?' This distinction is tracked as a readiness indicator for the microscopy activities in Lesson 2.

	to specific regions of the projected photograph and asks students to describe only what they SEE — not what they think it is.	seconds.] Cold-call 2 students. Write key observed details on the board in a two-column table: OBSERVATION POSSIBLE EXPLANATION. Explicitly say: 'We are filling the left column only right now — we will fill the right column as we learn more over the next lessons.'		
Explain Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: The FDA and Food Safety (PLIX) Source: CK-12  Search ARES for similar readings	The class constructs a large Know / Wonder / Notice (KWN) chart together on the front board. The teacher draws three columns: WHAT WE THINK WE KNOW WHAT WE NOTICE (from the photos) WHAT WE WONDER. Students contribute to each column in a structured round: first, every student writes their own KWN entries in their book (2 minutes); then contributions are gathered whole-class and the teacher scribes them on the board. Students are then introduced to the sub-strand Driving Question: 'Why did the ugali and chapati grow fuzzy patches overnight — and how can understanding the invisible organisms responsible help us protect our health and our food in Kenya?' The teacher reads it aloud and asks students to connect it to their 'Wonder' column entries: 'Which of your wonder questions feels like a piece of this big driving question?'	After populating the KWN chart from student contributions, explicitly narrate the scientific relevance: 'Everything in the KNOW column represents your starting mental model — it might be right, it might need refining. Everything in the WONDER column is the engine that will drive our next six lessons. In science, good questions are more valuable than quick answers.' Then reveal the driving question on a large poster or projected slide and read it aloud. Say: 'I want three volunteers to tell me — which ONE word in this driving question do you find most confusing or most interesting?' [WAIT 15 seconds.] Cold-call 3 students. Likely target words: 'invisible', 'organisms', 'protect'. For each word cold-called, ask the class: 'Does anyone want to add to that? Turn to your partner — 30 seconds.' Do NOT explain the words yet — record them on the board as 'words we will define together.' Finally, ask: 'On a scale of 1–3 with your fingers — 1 means I have no idea what is going on, 3 means I feel like I could start explaining this — show me where you are right now.' Scan the room. This is a quick confidence temperature-check, not a grade.	KWN CHART (Know / Wonder / Notice). The KWN chart is a metacognitive sensemaking tool that externalises the class's collective prior knowledge and makes it visible, revisable and comparable over time. Unlike a simple brainstorm, the separation of 'what we think we know' from 'what we notice' trains students to distinguish prior belief from direct evidence — a critical distinction in scientific reasoning. The chart remains posted throughout the sub-strand and is updated at the end of each lesson.	Observable indicator: Students' 'WONDER' entries reveal the quality of their scientific curiosity and the sophistication of their prior knowledge. The teacher looks for: (a) causal questions ('Why do different patches have different colours?') — indicates readiness for classification; (b) mechanistic questions ('How did the organism get onto the food if no one touched it?') — indicates readiness for mode-of-transmission content; (c) purely descriptive questions ('What is the name of the fuzzy thing?') — indicates students need more scaffolding to think causally. Teacher notes distribution across all three types to plan differentiation in Lesson 2.
Driving	Each student is given three sticky	Distribute sticky notes and say:	DRIVING QUESTION BOARD	Observable indicator: All students

Question Board (DQB) Creation  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: The FDA and Food Safety (PLIX) Source: CK-12  Search ARES for similar readings	notes (or tears three small pieces of paper if sticky notes are unavailable). On each note they write one question they have about the phenomenon — questions must be about the ugali/chapati patches specifically, not general science questions. Students stick their notes on the newly created DQB (a section of the front board or a large sheet of sugar paper). In pairs, students then do a 3-minute 'gallery walk' along the DQB, placing a tick on any note whose question they also want answered. The teacher facilitates a 3-minute whole-class cluster: similar questions are grouped into four emerging themes — (1) What are the patches? (2) How did they get there? (3) Are they dangerous? (4) Can we stop them? The teacher labels each cluster with a coloured marker and tells the class: 'Over the next lessons, we will answer each cluster. By the last lesson, every question on this board should have an answer written next to it — and you will be the ones writing those answers.'	'Write one genuine question per note — something you actually want to know. Your question must connect to these photographs. You have 2 minutes — go.' [Allow 2 minutes of silent writing.] Direct students to the DQB: 'Walk up, stick your notes anywhere on the board, then spend 2 minutes reading others' questions and ticking any you also want answered.' After the gallery walk, say: 'I am going to group these questions with you. Call out if you see two questions that are really asking the same thing.' Facilitate clustering with student input — do not impose categories, let them emerge. Once four clusters are visible, say: 'These four clusters are our roadmap for the sub-strand. In Lesson 2 we start answering Cluster 1 — What are the patches? — by looking at them far more closely.' Write 'Lesson 2 →' next to Cluster 1 in a different colour to signal the journey ahead. Photograph the DQB with a phone or camera so the record is preserved if notes fall off.	(DQB) — Student-Generated Question Clustering. The DQB is a public, persistent artefact that transforms student curiosity into a structured research agenda. By having students generate and then vote on questions, ownership of the inquiry shifts to the learners. The clustering step models the scientific practice of organising questions by theme (NGSS SEP 1), and the explicit connection of each cluster to a future lesson creates anticipation and a sense of coherent narrative across the sub-strand.	have contributed at least one note to the DQB. Quality check — the teacher scans for questions that are testable vs. purely philosophical. A testable question (e.g., 'Will the patches grow faster if we cover the food?') signals readiness for experimental design in the KICD-mandated project on mould growth. A non-testable question (e.g., 'Why does God let food go bad?') is acknowledged respectfully and redirected: 'That is a deep question — for today, let us focus on what we can investigate scientifically.'
Model Building Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: The FDA and Food Safety (PLIX) Source: CK-12	In the final 6 minutes, students individually draw a 'Before Model' — a simple annotated sketch in their exercise books showing their current best explanation of what is happening on the ugali and chapati. The sketch must include: (a) the food surface, (b) the patches (drawn as the student imagines them), (c) arrows showing where the patches came from, and (d) a label or sentence saying whether the patches are alive and why. Students are told: 'This is not a test — there is no right answer yet. This is a photograph of your thinking right	Announce the model-building task: 'I want you to draw your current best explanation — not what you wish you knew, but what you actually think right now. You have 5 minutes. Work in silence.' [Allow 5 minutes.] Circulate and observe — do not correct any model. In the final 1 minute, say: 'Turn to your partner and show them your model — not to judge it, just to see how differently two people can think about the same thing. You have 30 seconds.' Bring the class to attention: 'These models stay in your book. In our last lesson on microorganisms, I will ask you to	INITIAL ANCHOR MODEL (Pre-Instruction Mental Model). Drawing a pre-instruction model activates and makes visible students' existing mental models — including misconceptions — in a low-stakes format. Research in science education shows that students who draw and then revise models across a unit develop significantly deeper conceptual understanding than those who receive direct instruction alone. The 'Model 1 → Model 2' framing creates a built-in self-assessment opportunity and gives the teacher a rich artefact for tracking	Observable indicator: Every student has a labelled sketch with at least three components (food surface, patches, origin arrows). The teacher collects a rapid mental inventory of model types seen during circulation: (Type A) patches floating in from air with no mechanism — most common; (Type B) patches growing out of the food itself — indicates belief in spontaneous generation, a key misconception to address; (Type C) tiny organisms depicted landing on food and multiplying — most scientifically sophisticated. The distribution of model types informs

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D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) How many students attributed the patches to living organisms vs. non-living causes (dirt, chemicals, spontaneous rotting)? What does this distribution tell you about how much prior knowledge the class brings from Junior School Integrated Science? (2) Were students able to separate observation from inference during the Phenomena Gallery, or did they consistently conflate the two? If conflation was common, plan a 5-minute recap of the NOTICE / INFER / WONDER distinction at the opening of Lesson 2. (3) Review the DQB clusters — did a cluster emerge about health and disease (e.g., 'Can the patches make you sick?')? If so, this is an opportunity to foreground the KICD-mandated content on prevention and control of infections earlier in the sequence. (4) Did any student mention a personal experience of food spoilage at home (e.g., githeri left out, mandazi going stale)? These personal connections are gold — invite that student to share briefly at the start of Lesson 2 to reground the phenomenon in lived Kenyan experience. (5) Were there any students who appeared disengaged during the DQB construction? Consider assigning them the role of 'DQB Curator' — responsible for keeping the board tidy and photographed — as a structured participation strategy.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Photographs of leftover ugali and chapati stored overnight on a kitchen counter in Kisumu, showing white, green and black fuzzy patches, slimy areas and described sour smell. Students observed variation in patch colour, texture and location across the same food surfaces.
What did I learn?	Students surfaced prior knowledge about tiny organisms ('germs', 'bacteria') and began to distinguish between living and non-living causes of food change. They learned that scientists separate observations from inferences, and that the patches — whatever they are — appeared without anyone touching the food, meaning the source must be investigated systematically.
How does this explain the phenomenon?	No formal explanation was provided in Lesson 1 — this is deliberate. Students' initial mental models (Model 1) record their pre-instruction thinking. The Driving Question Board now holds the class's collective questions, organised into four clusters, which will be answered across Lessons 2–6 as students learn about types of microorganisms, modes of transmission, infections, prevention and economic importance.

LESSON 2 (40 minutes): Types of Microorganisms: Naming the Invisible Invaders on Our Ugali

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students will identify and classify the three major types of microorganisms that affect human beings — fungi, bacteria, and viruses — and use this classification to explain what is growing on the ugali and chapati in the phenomenon.
Knowledge	Students can name and describe the structural features of fungi, bacteria, and viruses; distinguish between them based on size, cell structure, and mode of reproduction; and correctly classify the fuzzy patches on the phenomenon food as fungal moulds.
Skills	Students practise guided research using digital devices or print media, interpret photomicrographs and scientific illustrations, organise information into a structured comparison table, and communicate findings using scientific vocabulary.
Attitudes	Students develop curiosity about the invisible world, appreciate the value of evidence-based classification over assumption, and build confidence in using scientific sources to answer questions that arise from everyday Kenyan life.
Key Inquiry Question	How are microorganisms important in day-to-day life? — addressed in this lesson by first establishing what microorganisms are and how they differ structurally, preparing students to later explore their roles in health, food, and the Kenyan economy.
Purpose in Storyline	This is the first Observe Phase lesson. In Lesson 1 students predicted what the patches might be; now they gather the first real evidence by researching microorganism types. The comparison table and photomicrograph analysis give students the conceptual vocabulary and structural knowledge they will need in Lessons 3 and 4 when they investigate transmission and infection. The lesson also resolves the Lesson 1 puzzle about whether all the patches are the same kind of organism — they are not, but today students confirm the fuzzy mould patches are fungal.
Safety Notes	If printed photographs of mouldy food are handled, students should not place them near their faces. If a teacher demonstration sample of actual mouldy food is used for visual reference only, it must remain sealed in a clear zip-lock bag and students must not open it. Hand hygiene should be observed after the lesson. No actual culturing of microorganisms occurs in this lesson.

B. LESSON OVERVIEW

Students begin with a quick revisit of their Lesson 1 predictions on the Driving Question Board (DQB), then transition into guided research using photomicrographs and print or digital resources to discover the three types of microorganisms — fungi, bacteria, and viruses. Using a structured worksheet, pairs construct a comparison table of structural features. The teacher uses Kenyan-context size-scale analogies to make the microscopic world concrete. The class then returns to the phenomenon photographs to classify the white, green, and black fuzzy patches as fungal mould, updating the DQB with new evidence and refining earlier predictions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Populations, communities, and	Students spend 4 minutes revisiting their Lesson 1 predictions displayed on the DQB. Each pair re-reads their sticky note	Teacher points to the DQB and says: 'Before we gather new evidence today, I want you to look at what we predicted in Lesson 1.'	Activating prior knowledge through targeted prediction anchored to the phenomenon. The focused question narrows the cognitive	Teacher circulates during the 90-second writing phase and scans notebooks to gauge whether students are drawing on biological

ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12 Search ARES for similar readings	prediction about what the patches are and whether they are alive. The teacher poses a focused warm-up question: 'From your prediction last lesson, write down in one sentence — do you think the white patch and the green patch on the ugali are the same type of organism or different types?' Students write their answer in their science notebooks before any discussion begins.	Do not change anything yet — we will do that at the end once we have evidence.' Teacher then poses the warm-up question verbally and writes it on the board: 'Same type of organism or different types — what do you think, and why?' Teacher instructs: 'Write your answer silently for 90 seconds. Do not discuss yet.' [WAIT TIME: 90 seconds of silent individual writing.] After writing, teacher selects three pairs at random using lollipop sticks to share their reasoning briefly. Teacher does not confirm or correct answers yet, saying: 'Hold that thought — our research today will help us decide.'	entry point from the broad Lesson 1 predictions to a specific classification question, preparing students to look for structural differences during research.	vocabulary (e.g., 'mould', 'bacteria', 'germs') or purely everyday language. This informs how much scaffolding to provide during the Observe Phase.
Observe Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12 Search ARES for similar readings	Students work in pairs for approximately 18 minutes. Each pair receives a Guided Research Worksheet containing: (1) a photomicrograph sheet with labelled images of Rhizopus stolonifer mould, Escherichia coli bacteria, and an influenza virus; (2) a blank three-column comparison table with rows for: type of cell (eukaryote / prokaryote / non-cellular), approximate size, presence of nucleus, cell wall material, mode of reproduction, and visible to naked eye; (3) two to three short reference paragraphs from a KICD-approved Grade 10 textbook or a printed summary from a curated source. Where digital devices are available, pairs may also use them to view additional photomicrographs. Students read, observe, and complete the comparison table together. Midway through the activity the teacher pauses the class to deliver the size-scale analogy: 'If a single grain of dry	Teacher distributes worksheets and says: 'Your job for the next 18 minutes is to become detectives. Use the photomicrographs and the reference text to fill in your comparison table. Focus on the evidence in front of you — not just what you think you remember.' Teacher circulates continuously, pausing at pairs to ask probing questions: 'What does the image tell you about whether this organism has a nucleus?', 'Where in the text did you find the size information — can you point to it?' At the 9-minute mark, teacher calls attention and delivers the Kisumu football pitch size-scale analogy as a 2-minute whole-class explanation with a quick sketch on the board showing the three organisms at relative scale. Teacher then says: 'Now go back to your table and make sure your size row reflects that scale.' [WAIT TIME after analogy: 30 seconds for students to process before returning to task.] Teacher uses	Guided inquiry with structured scaffolding — the comparison table prevents information overload and directs observation to scientifically significant structural features. The size-scale analogy uses a locally familiar landmark (a Kenyan school football pitch) to make abstract magnitude concrete and memorable. Pair work promotes collaborative sense-making and reduces anxiety about unfamiliar scientific vocabulary.	Teacher collects a quick show of completed comparison tables at the 16-minute mark by asking pairs to hold up their worksheet. Teacher scans for common errors — particularly students confusing bacteria (prokaryote) with fungi (eukaryote) — and notes these for the Explain Phase discussion. Students who have conflated viruses with bacteria are identified for targeted questioning.

	maize were the size of a typical bacterium, then to match the size of a human cell you would need to walk the entire length of a standard football pitch — such as the ones at Kisumu Boys High School grounds. A virus is so much smaller still that on the same scale it would be a single sesame seed next to that maize grain.' Students record this analogy in their notebooks next to their table.	targeted questioning for pairs who finish early: 'Look at the cell wall material row — fungi have chitin, plants have cellulose. Why might that difference matter for how we treat a fungal infection?'		
Explain Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12  Search ARES for similar readings	Whole-class discussion for 10 minutes. Teacher projects or draws a master comparison table on the board and co-constructs it with student contributions. Students verify and correct their own tables. The class then applies the classification to the phenomenon: teacher re-displays the phenomenon photographs (spoiled ugali and chapati) and asks students to use their new comparison table to answer: 'Which type of microorganism — fungus, bacterium, or virus — best explains the white, green, and black fuzzy patches we see here?' Students discuss in pairs for 60 seconds then share. The class reaches a consensus that the fuzzy patches are fungal moulds based on visible structure (multicellular, fuzzy hyphae visible to the eye, growing on organic food substrate). Teacher introduces the terms mould and spores, explaining that fungal spores were already present in the air of the kitchen and settled on the uncovered, moist, warm ugali overnight — a direct link back to the mystery of how the growth appeared without anyone touching the food.	Teacher opens the Explain Phase by saying: 'Let us build our class table together. I will not just give you the answers — I will ask you to justify each cell using evidence from your sources.' Teacher calls on different pairs for each row, asking: 'What is your evidence for that?' and 'Does anyone have a different answer — if so, show us where your source says that.' [WAIT TIME after each question: 10 seconds before accepting answers, to prevent the same fast responders from dominating.] When co-constructing the size row, teacher revisits the football pitch analogy: 'So if viruses are the smallest, why do you think we cannot see any viruses on the ugali photograph but we can see the fuzzy fungal patches?' After consensus on fungal mould classification, teacher says: 'Write this in your notebook as a scientific claim: The fuzzy patches on the ugali and chapati in the phenomenon are caused by fungal moulds. Your evidence is...' [WAIT TIME: 45 seconds for students to complete the evidence sentence themselves before teacher discusses it.] Teacher explicitly names the connection: 'This is new evidence. In Lesson 1 we did not	Co-construction of the master table makes student reasoning visible and correctable in real time. The claim-evidence sentence structure (adapted from science writing heuristic) trains students to separate conclusions from evidence. Returning to the phenomenon photographs keeps the abstract classification grounded in the original observable puzzle.	Teacher checks the claim-evidence sentences in notebooks by circulating during the 45-second writing pause. Any student who writes a claim without identifying structural evidence (e.g., writes only 'it is mould because it is fuzzy' without referencing eukaryotic cell structure or hyphae) is prompted: 'What from your comparison table supports that claim?' This distinguishes surface pattern-matching from structural understanding.

		know what the patches were. Now we have a classification. Later we will find out exactly how the spores got there.'		
Driving Question Board (DQB) Creation  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12  Search ARES for similar readings	Students spend 4 minutes updating the DQB. Each pair receives one new sticky note in a different colour from Lesson 1. They write one of the following — their choice: (a) a revision to their original prediction that they can now correct with evidence (e.g., changing 'they are all bacteria' to 'the fuzzy patches are fungi called moulds'), or (b) a new sub-question generated by today's learning (e.g., 'How did the mould spores get onto the ugali if no one touched it?' or 'If viruses are not cells, are they really alive?'). Pairs place their sticky note on the DQB. Teacher reads three aloud and explicitly links them to the storyline ahead: the spore dispersal question points to Lesson 3 on transmission, and the virus-alive question points to deeper classification work.	Teacher says: 'Scientists do not just accumulate facts — they update their thinking when they get new evidence. Go back to your original prediction sticky note. Can you now revise it, or has today raised a new question you did not have before?' Teacher distributes the new-colour sticky notes and gives 2 minutes of silent writing time. [WAIT TIME: 2 full minutes of silence — teacher does not prompt or hint during this time.] After pairs place notes, teacher reads three selected notes aloud and says: 'Notice how our questions are getting more specific. In Lesson 1 we asked what the patches were. Now we are asking how the spores travelled. That is the direction good science moves — from broad wondering to precise questions.' Teacher draws a visible arrow on the DQB from the original driving question toward the new sub-questions, labelling it 'Evidence from Lesson 2'.	The DQB update makes metacognitive growth visible — students can literally see their thinking becoming more sophisticated. Using a different sticky-note colour creates a visual timeline of evidence accumulation across the unit, reinforcing the idea that science is an iterative process of revision rather than a single correct answer.	The sticky notes are a written artefact of conceptual change. Teacher photographs the DQB after the lesson to track which students are revising predictions (evidence of knowledge integration) versus only adding new questions without revising earlier misconceptions. Students who have not revised a misconception (e.g., still claiming all patches are bacteria) are prioritised for a brief one-to-one check before Lesson 3.
Model Building Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12	In the final 4 minutes, students individually add one new annotated element to the rough model or sketch they began in Lesson 1 (or, if models were not started in Lesson 1, they draw a simple initial model now). The new element must show at least one of the three microorganism types with a label indicating its structural category (fungus-eukaryote, bacterium-prokaryote, virus-non-cellular) and an arrow indicating that mould spores were already present in the kitchen air before	Teacher says: 'Open your notebook to your Lesson 1 model or sketch. In the next 3 minutes, add to it using what you learned today. At minimum, label the fuzzy patches as fungal mould and add a small arrow showing where the spores came from. Use the scientific terms from your comparison table.' [WAIT TIME: 3 minutes of silent individual work — no pair discussion during model building to ensure individual accountability.] As the lesson closes, teacher says: 'Your model	Incremental model building across the unit externalises student understanding and makes conceptual growth tangible. Requiring annotation with scientific vocabulary from the comparison table reinforces disciplinary language use. The metaphor of the growing model mirroring the learning process to the phenomenon in a memorable way.	Teacher does a 60-second gallery walk during the last minute, visually scanning models for the two required elements — fungal mould label and spore-source arrow. Any model missing the spore arrow indicates the student has not yet connected the transmission mechanism to the phenomenon, which is the central conceptual target of Lesson 3. These students are noted for extra support at the start of the next lesson.

Search ARES for similar readings	landing on the food. Students do not need to complete a full model — this is an incremental addition that will be built upon in every subsequent lesson.	is not finished — it will grow every lesson just as the mould grew overnight on the ugali. By Lesson 8 it will tell the full story of the phenomenon.' Teacher previews Lesson 3: 'Next lesson we will investigate exactly how microorganisms travel from one place to another — because knowing how they move is the first step to stopping them.'		
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D. TEACHER REFLECTION

After the lesson, consider: Did students successfully distinguish fungi from bacteria using structural evidence, or did most rely only on visual appearance? Were the photomicrographs sufficiently clear — if students struggled to interpret them, sourcing higher-resolution images or a physical poster for Lesson 4 would help. Did the Kisumu football pitch size analogy land — did students reference it when discussing virus size, or did they revert to vague language like 'very small'? Were the DQB sticky notes showing genuine revision of misconceptions or simply restating the same ideas in new words? If the majority of students still conflate bacteria and fungi after this lesson, consider opening Lesson 3 with a 5-minute structured pair-compare of the two rows in the comparison table before moving to transmission. Also note whether digital device access was equitable — pairs without device access should have had equal quality print materials, and any gap in resource quality should be addressed before Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed photomicrographs and illustrations of fungi, bacteria, and viruses, and we examined photographs of the white, green, and black fuzzy patches growing on the ugali and chapati from our phenomenon.
What did I learn?	We learned that there are three major types of microorganisms — fungi (eukaryotic, with chitin cell walls), bacteria (prokaryotic, no true nucleus), and viruses (non-cellular, the smallest of the three) — and that the fuzzy patches on the ugali and chapati are caused by fungal moulds whose spores were already present in the kitchen air.
How does this explain the phenomenon?	We can now explain that the patches appeared without anyone touching the food because microscopic fungal spores, invisible to the naked eye, were floating in the kitchen air and settled on the warm, moist, uncovered ugali and chapati overnight, where they germinated and grew into the visible fuzzy colonies we see in the phenomenon photographs.

LESSON 3 (80 minutes (2 × 40-minute lessons)): Modes of Transmission: How Did the Mould Get onto the Ugali?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students use digital and print media to research, present, and evaluate the five main modes of transmission of microorganisms, then apply that understanding to explain precisely how mould spores were transmitted through the air onto the uncovered ugali and chapati in Kisumu.
Knowledge	Learners identify and explain the five modes of transmission of microorganisms in human beings: direct contact, respiratory droplets, contaminated water/food, vectors, and airborne transmission (KICD Sub-Strand 1.7b: explain modes of transmission of microorganisms in human beings).
Skills	Learners use digital or print media to research modes of transmission; organise and present findings to peers; construct a cause-and-effect chain linking airborne transmission to the anchoring phenomenon; and update the Driving Question Board with new evidence (KICD suggested learning experience: 'use digital or print media to search for information on the modes of transmission of microorganisms in human beings').
Attitudes	Learners appreciate how understanding transmission routes empowers communities — including families in Kisumu — to make informed decisions about food storage, hygiene, and disease prevention, reflecting KICD value of Responsibility and the National Goal of promoting positive attitudes towards good health.
Key Inquiry Question	How are microorganisms important in day-to-day life?
Purpose in Storyline	Lesson 3 is the second Observe phase in the 8-lesson sequence. Lessons 1–2 established that microorganisms exist and are classified into types (fungi, bacteria, viruses). Lesson 3 now answers the HOW of spread: students gather and share evidence on transmission routes, then converge on airborne transmission as the specific mechanism that explains the fuzzy patches on the ugali and chapati. This evidence will be needed in Lesson 4 (infections and disease) and Lesson 5 (prevention and control), making it a critical bridging lesson on the DQB storyline.
Safety Notes	No live cultures are handled in this lesson. If physical print materials include images of infected wounds or diseased tissue, preview with students that some images may be graphic and allow any learner to use the text-only version. Ensure any shared digital devices are cleaned before and after use (wipe screens/keyboards). Remind students to wash hands before and after handling any food-related print/photo materials. If a student has a known respiratory condition, seat them away from any dusty printed materials.

B. LESSON OVERVIEW

Students open by revisiting the anchoring phenomenon photograph (spoiled ugali and chapati from Kisumu) and the Driving Question Board from Lessons 1–2. They make a brief written prediction about HOW the mould got onto the food before any instruction occurs (Predict Phase). In the Observe Phase, five expert groups — each assigned one transmission route (airborne, direct contact, respiratory droplets, contaminated water/food, vectors) — research their route using digital devices or print resource cards, then deliver a structured 3-minute Gallery Walk presentation. The Explain Phase uses a Cause-and-Effect Chain graphic organiser to connect airborne transmission directly to the phenomenon. Groups then reconvene around the DQB to post new evidence cards and revise their cumulative model of microorganism spread. The lesson closes with a 3-2-1 formative exit card.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Graphs of Linear Models of Direct Variation (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner opens their science notebook and studies the projected phenomenon photograph: the sufuria of ugali and half-eaten chapati covered in white, green, and black fuzzy patches. They silently re-read the driving question displayed on the board. Without prompting from the teacher, each learner writes a 2–3 sentence prediction responding to: 'The kitchen looked clean and no insects were seen. How do you think the mould patches got onto the ugali and chapati overnight?' Learners then share predictions with a shoulder partner for 90 seconds before two or three predictions are shared aloud with the class. Predictions are NOT corrected at this stage — they are pinned to a 'Before Evidence' column on the DQB.	Display the phenomenon photograph on the board or ARES screen. Say verbatim: 'Look carefully at this ugali and chapati from a kitchen in Kisumu. We know from our last two lessons that fungi, bacteria, and viruses exist. Today's question is different — it is about TRAVEL. Before I teach you anything new, write in your notebook: how did those mould patches get onto the food? The kitchen was clean. No insects. No one touched it. Think silently for 2 minutes.' [SET TIMER — 2 minutes silent individual writing. Do not circulate or give hints.] After 2 minutes: 'Share your prediction with your shoulder partner. You have 90 seconds.' [WAIT — 90 seconds.] Cold-call 3 learners using lolly-stick or name cards — choose one learner who looks confident, one who looks uncertain, one from the back row. For each, ask: 'What is your prediction and what makes you think that?' Do NOT affirm or correct. Say: 'These are all interesting ideas. Let us gather evidence today to test them.' Pin 3 representative prediction cards to the DQB 'Before Evidence' column.	Strategy: Anchored Prediction. By requiring a written prediction tied specifically to the phenomenon BEFORE new content, this activates prior knowledge from Lessons 1–2 (microorganism types) and creates cognitive dissonance when evidence challenges their ideas. Research shows predictions increase retention of corrective evidence by forcing learners to compare their mental model against new data.	Teacher scans notebooks during the 2-minute silent write. Observable indicator: learner has written at least one causal mechanism (e.g., 'through the air', 'from someone's hands', 'from water', 'from a fly') rather than simply restating the observation. If more than half the class writes only 'the mould grew there' with no mechanism, pause and prompt: 'Think about HOW a mould spore could move from one place to another — what could carry it?'
Observe Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES	The class is divided into five Expert Groups of approximately 5–6 learners each. Each group receives an Assignment Card naming their transmission route and a Resource Pack (printed ARES resource cards, or access to a digital ARES offline module on that route). The five routes are:	Distribute Assignment Cards and Resource Packs. Say: 'Each group is now an expert on one way microorganisms travel. You have 18 minutes to research your route, fill in your Recording Sheet, and prepare a 3-minute presentation. Every person in your group must have a speaking part — decide	Strategy: Jigsaw Expert Research + Gallery Walk. The Jigsaw structure ensures every learner becomes a genuine expert in one domain, creating authentic purpose for the presentation. The Gallery Walk distributes the cognitive load of learning all five routes across the group rather	During circulation, teacher checks Recording Sheets for the 'Could this explain the ugali phenomenon?' field. Observable indicator: learner can articulate at least one reason why their route could OR could not explain the phenomenon with a logical justification (e.g., 'Direct contact

for similar videos  HTML: Graphs of Linear Models of Direct Variation (Flexbook) Source: CK-12  Search ARES for similar readings	Group A — Airborne Transmission; Group B — Direct Contact Transmission; Group C — Respiratory Droplet Transmission; Group D — Contaminated Water and Food Transmission; Group E — Vector Transmission. Each group has 18 minutes to: (1) read/watch their resource material, (2) complete a structured Research Recording Sheet with four fields — Definition, Kenyan Example of a disease spread this way, How this route could or could NOT explain the Kisumu ugali phenomenon, One prevention method. Groups then prepare a 3-minute spoken presentation on a single A3 sheet or mini-whiteboard to share during the Gallery Walk. Each group member must be assigned a speaking role.	your roles now.' [Start timer — 18 minutes.] Circulate in a structured pattern: visit each group at least twice. At each visit, ask one targeted question: Group A (Airborne): 'Can you name a specific particle that travels by air — what is it called in fungi?' Group B (Direct Contact): 'Give me a Kenyan example — think about shared utensils at a nyama choma stand.' Group C (Droplets): 'What is the difference between a droplet and an airborne particle? Size matters.' Group D (Food/Water): 'Think about water from an uncovered well near Lake Victoria — what might be in it?' Group E (Vectors): 'Name a vector common in the Rift Valley region.' [At 15 minutes, give a 3-minute warning.] Then transition to Gallery Walk: 'Post your A3 sheet on the wall. Groups will rotate every 3 minutes. One person from each group stays at their poster as a docent to answer questions. The rest of the class rotates. Take notes on your Recording Sheet for every route — you will need this for the next phase.' [Run 5 rotations × 3 minutes = 15 minutes. Ring bell or clap signal at each rotation.]	than through direct instruction alone. Connecting each route back to the Kisumu phenomenon on the Recording Sheet prevents the research from becoming abstract — every fact must be tested against the specific evidence from the anchoring phenomenon.	could not explain it because no one touched the ugali overnight'). Teacher notes any group that conflates airborne with droplet transmission — this is a key misconception to address in Phase 3. Post Gallery Walk: teacher quickly scans 3–4 notebooks for notes from all five stations — if a learner has blank sections, pair them with a note-comparing partner before Phase 3 begins.
Explain Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Graphs of Linear Models of Direct Variation	All learners return to their seats. The teacher projects a blank Cause-and-Effect Chain graphic organiser (5 boxes connected by arrows) on the board. Working first individually (4 minutes), then in table groups of mixed expertise (4 minutes), learners complete the chain to answer: 'Starting from a mould fungus somewhere in the environment, trace the exact path that led to the fuzzy patches on the ugali in the Kisumu kitchen.' The chain must include: (1) Source —	Project the blank Cause-and-Effect Chain. Say: 'You have just become experts on five transmission routes. Now use that evidence to solve our original puzzle. Individually, fill in the five boxes of this chain to trace exactly how the mould got from the environment onto the ugali. You have 4 minutes — silent, individual.' [WAIT — 4 minutes. Do not circulate during individual think time.] 'Now compare your chain with your table group. Agree on one best chain. 3	Strategy: Cause-and-Effect Chain with Phenomenon Anchoring. The graphic organiser forces learners to sequence mechanistic reasoning (not just list facts) and to connect abstract transmission concepts to the specific, visible evidence from the phenomenon. The two challenge questions use 'controlled variable' thinking (covering the food; vectors) to deepen causal understanding and prevent surface-level pattern matching. This mirrors NGSS	Teacher collects or photographs 4–5 completed individual chains before group comparison. Observable indicator of meeting expectations: all five boxes completed with scientifically accurate content AND box 2 correctly identifies airborne transmission as the primary mechanism with a justification referencing spore size/air currents. Indicator of exceeding expectations: learner spontaneously notes that multiple

(Flexbook) Source: CK-12 Search ARES for similar readings	where the spores originated; (2) Transmission mechanism — which mode and why airborne fits; (3) Environmental conditions — why overnight on an uncovered sufuria; (4) Colonisation — what the spores did when they landed; (5) Visible outcome — white, green, black fuzzy patches plus sour smell. Groups share their chains; class builds a consensus chain on the board. Teacher then poses two challenge questions for cold-call response.	minutes.' [WAIT — 3 minutes.] Cold-call Group A (Airborne experts) first: 'Walk us through your chain.' After they present, ask the class: 'Do we agree that airborne transmission is the best explanation here? Hands up if you agree — now hands up if you disagree or want to add something.' [WAIT 10 seconds after each hand-raise prompt.] Cold-call one dissenting voice if any. Then pose Challenge Question 1: 'If the chapati had been covered with a clean kanga cloth, would the mould still have appeared? Explain using transmission mode vocabulary.' [WAIT 15 seconds — full silence.] Cold-call 2 learners. Challenge Question 2: 'Could vector transmission also have played a role here, even though no insects were visible? Defend your answer.' [WAIT 15 seconds.] Cold-call 2 different learners. Consolidate on the board: draw the agreed consensus chain. Say: 'Airborne transmission — specifically the movement of fungal spores through the air — is the mechanism that best fits our evidence. Spores are microscopic, they are produced in enormous numbers by fungi, they are light enough to remain suspended in air currents, and they land on nutrient-rich surfaces like ugali and germinate rapidly in warm, humid kitchen conditions typical of a Kisumu home overnight.'	Science and Engineering Practice 6: Constructing Explanations.	routes (e.g., airborne + contaminated surface from a previous handling) could interact. Misconception to flag: any chain that lists 'dirty air' as a cause without specifying spores — teacher addresses this immediately: 'What specific structure of the fungus travels in the air? We met this in Lesson 1.'
Driving Question Board (DQB) Creation  VIDEO: Populations.	Each learner writes two DQB contributions on separate sticky notes or index cards: (1) an EVIDENCE card (green) — one fact about transmission they are now confident about based on	Distribute green and orange cards. Say: 'Before we close, you are going to update our Driving Question Board. On the GREEN card, write one specific piece of evidence from today that helps	Strategy: Driving Question Board as a Living Knowledge Wall. The DQB functions as a visible, cumulative model of the class's growing understanding. Separating 'what we know' from 'what we still	Teacher scans all green cards during the posting phase. Observable indicator: green card contains a specific claim (e.g., 'Airborne transmission means fungal spores float in the air and

communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Graphs of Linear Models of Direct Variation (Flexbook) Source: CK-12 Search ARES for similar readings	today's research; (2) a NEW QUESTION card (orange) — one new question today's lesson has raised that is not yet answered. Learners post their cards on the class DQB in the appropriate columns: 'What we now know' and 'What we still wonder'. The class then does a 3-minute silent gallery read of all new cards. Teacher highlights two or three 'still wonder' cards aloud and tells students which upcoming lesson will address them.	answer our driving question about the ugali. Use the words: transmission, spore, airborne, or another transmission vocabulary word. On the ORANGE card, write one question you now have that today's lesson did NOT answer — think about what you still need to know to fully explain the phenomenon.' [WAIT — 3 minutes writing time.] 'Post your cards now.' [Allow 1 minute for posting.] 'Stand and do a 3-minute silent read of what your classmates have posted — look for a green card that surprises you and an orange card that matches your own question.' [WAIT — 3 minutes.] Teacher reads aloud 2 green cards: 'I want to highlight these two evidence cards — they show strong use of today's vocabulary.' Read 2 orange cards: 'These two questions — about how we stop transmission and about what infections these microorganisms cause — will be answered in Lessons 4 and 5. Keep them in mind.'	wonder' develops metacognitive awareness — learners must distinguish established evidence from open inquiry, a key scientific habit of mind aligned with NGSS Science and Engineering Practice 1 (Asking Questions) and Practice 8 (Obtaining, Evaluating, and Communicating Information).	land on food') rather than a vague statement ('microorganisms spread'). Teacher collects all green and orange cards at the end of the lesson for review — cards with vague language are individually followed up in Lesson 4's opening.
Model Building Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Graphs of Linear Models of Direct Variation (Flexbook) Source: CK-12	Learners retrieve their cumulative 'Microorganism Spread Model' diagram begun in Lesson 1 and updated in Lesson 2. Using a different colour pen (to show the revision layer from Lesson 3), each learner adds: (1) arrows labelled with the five transmission routes pointing toward a central 'Host' figure; (2) a highlighted arrow for AIRBORNE transmission leading specifically to the ugali/chapati icon on their model; (3) a brief annotation (1–2 sentences) explaining why airborne transmission best fits the Kisumu phenomenon. Learners share their updated models with a	Say: 'Take out your cumulative model from Lessons 1 and 2. Use your new colour pen — this colour represents the evidence layer from Lesson 3. Add the five transmission routes as labelled arrows. Then highlight the airborne arrow and write next to it why it explains our ugali phenomenon. You have 6 minutes.' [WAIT — 6 minutes. Circulate and look for models that are missing transmission-route labels or that have placed the airborne arrow incorrectly.] 'Now share your model with your partner using this sentence frame: I can see in your model that... and one thing to	Strategy: Iterative Model Revision with Colour-Coded Evidence Layers. Using a new colour for each lesson's additions makes the growth of understanding visible and tangible. Students can literally see which ideas came from which evidence sessions, reinforcing the NGSS Science and Engineering Practice 2 (Developing and Using Models) and metacognitive reflection. The peer feedback sentence frame scaffolds constructive scientific communication without reducing it to praise or criticism.	Observable indicator of meeting expectations: updated model shows all five transmission routes as labelled arrows AND includes a highlighted airborne arrow with a written annotation that mentions fungal spores and the uncovered food surface. Observable indicator of approaching expectations: model shows three or more routes but the airborne annotation is missing or vague. Teacher uses the 5 collected models to identify any persistent misconceptions (e.g., conflating droplet with airborne) and opens Lesson 4 with a targeted clarifying question based on what is seen.

Search ARES for similar readings	partner for peer feedback using one sentence: 'I can see in your model that... and one thing to make it clearer would be...'	make it clearer would be...' [WAIT — 2 minutes peer feedback.] Close the lesson: 'Every lesson, your model grows. Today you added the HOW of microorganism travel. In Lesson 4, you will add the WHAT — what infections these transmission routes cause, including in your own community here in Kenya. Keep your model safe.' Collect 5 models at random to review overnight — return with written feedback before Lesson 4.		
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D. TEACHER REFLECTION

After the lesson, review the 5 randomly collected cumulative models and all DQB cards before Lesson 4. Ask yourself: (1) Did the majority of learners correctly identify airborne transmission — not direct contact or vectors — as the primary mechanism for the ugali phenomenon? If not, plan a 5-minute re-engagement at the start of Lesson 4 using a think-pair-share around the question: 'A covered sufuria versus an uncovered sufuria — what difference would covering make, and which transmission route does this target?' (2) Were the orange 'still wondering' DQB cards asking questions that point forward into Lessons 4–5 (infections, prevention), or are learners still confused about Lesson 3 content (the transmission routes themselves)? If more than 30% of orange cards show confusion about Lesson 3 content, build a 10-minute consolidation into the opening of Lesson 4. (3) Did the Gallery Walk run smoothly — did every group actually rotate and take notes, or did some groups linger? If rotation was problematic, consider a teacher-facilitated jigsaw report-back (each expert group presents to the full class in sequence) as an alternative for the next research lesson. (4) Reflect on equity: did female learners and learners who are typically quieter get cold-called equitably? Review your lolly-stick records. (5) Consider: is there a local health clinic, community health volunteer (CHV), or public health officer near the school who could visit in Lesson 5 to speak about transmission and prevention from a real Kenyan public health perspective? Begin making that arrangement now.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students researched five modes of transmission (airborne, direct contact, respiratory droplets, contaminated water/food, vectors) using digital or print media resource packs, presented findings in a Gallery Walk format, and gathered evidence on how microorganisms travel from source to host in Kenyan contexts.
What did I learn?	Airborne transmission — the movement of fungal spores suspended in air currents — is the specific mechanism that best explains how mould appeared on the uncovered ugali and chapati in the Kisumu kitchen overnight. Different transmission routes determine which prevention and control methods are most effective.
How does this explain the phenomenon?	Students constructed a Cause-and-Effect Chain tracing the path of mould spores from an environmental source, through the air, onto the nutrient-rich surface of warm ugali left uncovered in a Kisumu kitchen, producing the white, green, and black fuzzy patches and sour smell observed in the anchoring phenomenon.

LESSON 4 (80 minutes): Infections Caused by Microorganisms: Fungi, Bacteria, and Viruses

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to identify types of infections caused by microorganisms in human beings and link each infection to its mode of transmission, thereby advancing their explanation of why the fuzzy patches appeared on the ugali and chapati in Kisumu and what risks those same organisms pose to human health.
Knowledge	Learners will be able to: (a) identify types of infections caused by microorganisms in human beings — fungal (ringworm, oral thrush), bacterial (cholera, typhoid, tuberculosis), and viral (influenza, HIV, COVID-19); (b) describe the causative microorganism and key symptoms for each infection; (c) explain how the mode of transmission of each microorganism determines the infection it causes.
Skills	Learners will: use structured discussion and jigsaw reading to gather and synthesise evidence; construct a comparison table linking infection type, causative organism, mode of transmission, and body system affected; update the Driving Question Board by revising earlier predictions and posting new sub-questions; apply NGSS Science and Engineering Practice 6 (Constructing Explanations) to connect transmission routes from Lesson 3 to specific infections.
Attitudes	Learners will appreciate the effect of microorganisms in day-to-day life; show respect when listening to peers sharing sensitive health information (e.g., HIV/TB); demonstrate responsibility by committing to behaviour changes that reduce infection risk in their homes and communities.
Key Inquiry Question	How are microorganisms important in day-to-day life?
Purpose in Storyline	In Lesson 3, learners investigated modes of transmission — airborne droplets, contaminated water and food, direct skin contact, and vector-borne routes. Lesson 4 is the first Explain Phase: learners now name the specific diseases those transmission routes produce, classify them by causative organism type (fungus, bacterium, virus), and begin to construct a mechanistic explanation of the phenomenon. The Lake Victoria cholera context grounds bacterial infections in lived Kenyan experience, while the mould patches on the ugali and chapati serve as the entry point for understanding fungal infections of the body. By the end of this lesson, learners have enough explanatory vocabulary to start answering the driving question.
Safety Notes	No live cultures or food samples are handled in this lesson. All evidence comes from photographs, information cards, and discussion. Remind learners that HIV/TB discussions require sensitivity and confidentiality; no learner should be asked to disclose personal or family health information. Printed infection cards should be handled with clean hands and disposed of hygienically.

B. LESSON OVERVIEW

Lesson 4 is the first full Explain Phase in the eight-lesson sequence. Building on the modes of transmission mapped in Lesson 3, learners now investigate the specific infections caused by fungi, bacteria, and viruses. The lesson opens by returning to the photograph of the mouldy ugali and chapati from Kisumu and asking: 'If those same fungi were on your skin or in your mouth, what would happen?' A structured Predict-and-Share routine surfaces prior knowledge and misconceptions. The core activity is a Jigsaw — three expert groups each research one infection category (fungal, bacterial, viral) using information cards and short reading texts embedded in ARES, then teach their peers. During the Explain phase, a whole-class discussion integrates findings into a master comparison table. The Driving Question Board (DQB) is updated: learners physically move or revise sticky-note predictions from Lessons 1–3 and add new sub-questions. The lesson closes with a Model-Building prompt in which learners annotate a body-outline diagram to show where each infection type acts. Throughout, the Lake Victoria cholera outbreak context and the TB prevalence in informal settlements near Nairobi ground bacterial infections in authentic Kenyan public-health realities.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Sexually Transmitted Infections (Flexbook) Source: CK-12  Search ARES for similar readings	Learners view the same opening photograph of the mouldy ugali and chapati on a kitchen counter in Kisumu that anchored Lesson 1. They then see a second paired image: a child's arm showing ringworm (tinea corporis) and a close-up of white oral thrush patches on a tongue. Working individually for 3 minutes, learners respond in their ARES notebooks to three written prompts: (1) 'What type of organism do you think causes ringworm — is it the same kind as the mould on the ugali? Give a reason.' (2) 'Cholera outbreaks have been reported near Lake Victoria. What kind of microorganism do you think causes cholera, and how do you think it reaches the human body?' (3) 'Name one viral infection you have heard of. How do you think it spreads?' After writing, learners share with a shoulder partner for 2 minutes (Think-Pair). Three volunteers are cold-called to share their pair's ideas aloud. Teacher records key prediction phrases verbatim on the DQB under the heading 'What we think we know — Lesson 4 start'.	Display both images on the ARES screen. Read each prompt aloud clearly. Say: 'You have 3 minutes to write your own ideas — no wrong answers at this stage, we are building on what we already know from Lessons 1 to 3.' Set a visible 3-minute timer. WAIT in silence. After the timer, say: 'Turn to your shoulder partner and share what you wrote. You each have 1 minute.' After 2 minutes, cold-call exactly 3 pairs using a random name selector or row order: 'Tell me one idea your pair agreed on.' Record each response verbatim on a sticky note on the DQB — do not correct yet. Say: 'Hold those ideas — by the end of today you will be able to test each one against evidence.'	Think-Pair-Share (prediction activation): surfaces existing mental models about infection causation before instruction, creating cognitive dissonance that the jigsaw evidence will resolve. Recording predictions verbatim on the DQB makes prior knowledge visible and revisable — a key feature of NGSS Practice 6 (Constructing Explanations).	Teacher scans written notebook responses during the 3-minute individual writing time — look for: (a) correct linking of fungi to both food mould and skin ringworm; (b) mention of contaminated water as a cholera route (evidence from Lesson 3); (c) naming of any viral disease. Misconceptions to flag: 'ringworm is caused by a worm' (very common); 'cholera comes from eating bad ugali' (partial — conflates food spoilage fungus with bacteria in water). Note these on a clipboard for targeted correction during the Explain phase.
Observe Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners are divided into three numbered Expert Groups of roughly equal size (approximately 10–12 learners each in a class of 35). Each group receives a dedicated ARES information card set and a guiding question sheet: – Group 1 — FUNGAL INFECTIONS: Information	Distribute pre-printed ARES information card sets (or direct learners to the ARES offline digital card viewer). Say: 'You are now infection experts. Your job is to understand your category deeply enough to teach the rest of the class. You have 15 minutes. Use the four-column Evidence Table on your worksheet.' Circulate to each group at least twice. At Group 2,	Jigsaw Cooperative Learning: by making each learner the expert-teacher for one infection category, this strategy deepens individual accountability and requires learners to process information at a level of understanding sufficient to explain it to others (Bloom's: Apply/Analyse). The four-column Evidence Table is a structured data-recording tool that mirrors the	Circulate and check Evidence Tables for accuracy in the 'Mode of Transmission' column — this is the key cross-lesson link. Observable success indicator: learners correctly write 'airborne droplets' for TB and influenza, 'contaminated water/food' for cholera and typhoid, 'direct skin contact / fomites' for ringworm, and 'bodily fluids' for HIV. Flag any

 HTML: Sexually Transmitted Infections (Flexbook) Source: CK-12  Search ARES for similar readings	cards on ringworm (tinea corporis) and oral thrush (candidiasis). Cards include: causative organism name (Trichophyton spp. for ringworm; Candida albicans for thrush), body parts affected, typical symptoms, photograph, and a short paragraph on how the same fungal kingdom that produces the green mould on ugali can infect human skin and mucous membranes. A local context note describes how ringworm is common in primary schools in Kisumu and Mombasa, often spread by sharing combs and towels. – Group 2 — BACTERIAL INFECTIONS: Information cards on cholera (Vibrio cholerae — contaminated Lake Victoria water supply link), typhoid (Salmonella typhi — contaminated water and food), and tuberculosis / TB (Mycobacterium tuberculosis — airborne droplets in crowded settings such as Nairobi Eastlands). Each card states: causative bacterium, mode of transmission, key symptoms, and one Kenyan public-health statistic (e.g., 'Kenya reported over 28,000 TB notifications in 2022 — WHO Kenya Country Profile'). – Group 3 — VIRAL INFECTIONS: Information cards on influenza (seasonal flu — droplet transmission), HIV (body fluids — sexual contact, shared needles,	prompt: 'Look at the cholera card — what does it say about the water source? How does that connect to what we learned about waterborne transmission in Lesson 3?' WAIT 10 seconds for a learner response before offering a hint. At Group 1, challenge the ringworm-is-a-worm misconception if it surfaces: 'What kingdom does the causative organism belong to? Is that the same kingdom as the mould on the ugali?' At Group 3, prompt sensitivity: 'How should we talk about HIV respectfully in our community?' After 15 minutes, give a 2-minute warning. Then reshuffle learners into Jigsaw Teaching Groups of 3 (one expert from each original group per new group) — approximately 11–12 new groups. Jigsaw Teaching Groups spend 12 minutes: each expert teaches their category to the other two, who fill in their own Evidence Table for all three infection types. Teacher circulates and listens for accuracy, correcting errors quietly by asking guiding questions rather than giving answers.	NGSS Practice 4 (Analysing and Interpreting Data) scaffold — learners are not just reading but categorising and linking evidence. The explicit Lesson 3 vocabulary requirement (modes of transmission) forces integration across lessons, advancing the storyline.	group that writes 'bad air' or 'dirt' as transmission routes — these indicate that Lesson 3 learning has not consolidated and will need reinforcement during the Explain phase whole-class discussion.
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	mother-to-child), and COVID-19 (SARS-CoV-2 — airborne and droplet). Cards include Kenyan context: the 2020–2022 COVID-19 response in Nairobi, and HIV prevalence statistics for Kisumu County (one of Kenya's highest-prevalence counties). Each Expert Group spends 15 minutes reading and discussing their cards. They must complete a Group Evidence Table with four columns: Infection Name Causative Organism (type + scientific name) Mode of Transmission (from Lesson 3 vocabulary) Main Body System / Site Affected. Groups appoint a recorder and a spokesperson.			
Explain Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Sexually Transmitted Infections (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to a whole-class configuration. The teacher displays an empty Master Comparison Table on the ARES shared screen with columns: Infection Type of Microorganism Causative Organism Mode of Transmission Body Site / System Affected Link to Ugali/Chapati Phenomenon (Yes/No/Explain). One spokesperson from each original Expert Group (Fungal, Bacterial, Viral) presents their category's findings in 3 minutes each. After each presentation, the class completes that section of the Master Table in their notebooks. The teacher then facilitates a 10-minute structured discussion using four anchor questions: (1) 'Ringworm and the mould on the ugali are both fungi. What is similar and what is different about how they cause harm?' (2) 'Cholera outbreaks keep	Project the empty Master Table. Invite the Fungal spokesperson: 'Group 1, you have 3 minutes — tell us the infection name, the causative organism, how it spreads, and where it attacks the body.' After presentation, say: 'Everyone — fill in the Fungal row on your table now. You have 90 seconds.' Repeat for Bacterial and Viral spokespeople. Then launch structured discussion. For Question 1, say: 'I want you to think about this for 12 seconds before anyone speaks.' Start timer visibly. Cold-call two learners by name. For Question 2, draw attention to the Lake Victoria photograph displayed on screen: 'Look at this image of fishing communities on the lake shore. What do you see that might increase cholera transmission risk?' WAIT 12 seconds. Cold-call 2 learners. For the 'Explanation So	Structured Academic Controversy discussion + Anchor Questions: the four anchor questions are deliberately designed to create productive cognitive conflict (e.g., 'all fungi are the same' vs. 'fungi can infect food AND human skin in different ways'). The sentence frame 'The evidence from my card shows that... and this connects to the phenomenon because...' is an NGSS Practice 6 scaffold that forces learners to ground every explanatory claim in evidence and to link it back to the anchoring phenomenon — the mouldy ugali and chapati. The Master Comparison Table serves as a shared, negotiated record of the class's collective understanding at this point in the storyline.	Collect and scan the 3-sentence 'Explanation So Far' entries. Success criteria: (1) correct infection-organism pairing (e.g., cholera → <i>Vibrio cholerae</i>, not just 'bacteria'); (2) mode of transmission stated explicitly; (3) a named Kenyan location or context. Common errors to address next lesson: conflating bacterial and viral diseases; omitting transmission route; writing 'germs' rather than specific organism type. These entries inform the differentiation plan for Lesson 5 (Prevention and Control).

	happening near Lake Victoria. Based on what you now know, why is the water transmission route particularly dangerous in that region?' (3) 'TB spreads through airborne droplets in crowded places. Where in Kenya would you expect TB risk to be highest — and why?' (4) 'COVID-19, influenza, and HIV are all viral. Does that mean they spread the same way? Explain.' For each question, teacher uses a cold-call of 2 learners after 12 seconds of WAIT TIME, then opens for voluntary responses. Learners are expected to use the sentence frame: 'The evidence from my card shows that... and this connects to the phenomenon because...'. The session closes with learners writing a 3-sentence 'Explanation So Far' in their notebooks: one sentence for fungal infections, one for bacterial, one for viral — each sentence must mention the mode of transmission and at least one Kenyan context.	Far' writing, say: 'Write three sentences in your notebook. Each must name a real Kenyan place or context. You have 4 minutes.' Circulate and read over shoulders — give a 'thumbs up' to correct entries, tap the notebook gently for entries needing revision and whisper: 'Does your sentence say HOW it gets into the body?'		
Driving Question Board (DQB) Creation  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Sexually Transmitted Infections (Flexbook) Source: CK-12	Learners return to the physical or digital DQB that has been accumulating since Lesson 1. The teacher projects the DQB and reads aloud the predictions posted at the start of today's lesson. Working in pairs, learners are given two tasks over 8 minutes: (1) REVISE — each pair identifies one sticky note from Lessons 1–3 that they can now correct or refine based on today's evidence. They write a revised version on a green sticky note and physically replace or annotate the original. Example: a Lesson 1 prediction 'the patches are bacteria' may now be revised to 'the white and green fuzzy patches on the ugali are fungi	Point to the DQB: 'Look at the predictions we made at the start of today and in earlier lessons. You now have new evidence. Pair up and do two things: find one prediction you can improve, and write one new question this lesson has sparked.' Distribute green sticky notes (revisions) and orange sticky notes (new questions). Set a 7-minute timer. Circulate — prompt pairs who are stuck: 'What did you learn today about ringworm that you didn't know at the start of Lesson 1? Can any of the old predictions be made more precise?' After 7 minutes, say: 'Stand up and do a 3-minute gallery walk — read your	Driving Question Board revision: the DQB is a living epistemic artefact that externalises the class's growing understanding across the 8-lesson sequence. The green/orange sticky-note protocol distinguishes between revised knowledge claims (green) and open inquiry questions (orange), mirroring the scientific practice of distinguishing what is known from what remains uncertain — a core feature of NGSS Practice 1 (Asking Questions). Requiring learners to physically revise earlier predictions rather than simply add new ones reinforces the idea that science explanations are revisable in light	Teacher photographs the DQB at the end of this phase (for ARES digital portfolio). Observe: (1) Are revised predictions more specific and evidence-grounded than originals? (2) Do new questions reflect genuine conceptual advance (e.g., asking about mechanisms of infection, immunity, or prevention) rather than surface curiosity? (3) Are any learners unable to revise a prediction — indicating they did not consolidate the jigsaw content? These learners are identified for a brief check-in at the start of Lesson 5.

 Search ARES for similar readings	(moulds), not bacteria — bacteria can also spoil food but they look slimy and smell different.' (2) NEW QUESTIONS — each pair writes one new sub-question that today's lesson has raised, on an orange sticky note. Examples from anticipated learner responses: 'If cholera is in Lake Victoria water, why don't all fishermen get cholera every day?'; 'Can the Candida fungus that causes oral thrush grow on ugali too?'; 'Why does TB only infect the lungs and not the whole body?'. Pairs place new questions on the DQB under the section 'What we still want to know'. The teacher facilitates a 3-minute gallery walk around the DQB, then highlights two or three of the most generative new questions aloud: 'These questions will drive our next lesson on prevention and control — hold onto them.'	classmates' new questions.' After the gallery walk, read aloud two high-quality new questions and say: 'These are exactly the kinds of questions scientists ask. We will use them in Lesson 5.'	of new evidence.	
Model Building Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Sexually Transmitted Infections (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner opens a pre-printed or ARES-displayed outline of a human body silhouette (front view). Using colour-coded annotation (green for fungal, red for bacterial, blue for viral), learners spend 8 minutes labelling the body outline to show: (a) where on or in the body each infection acts (e.g., skin — ringworm; lungs — TB and influenza; intestines — cholera and typhoid; mouth — oral thrush; immune system — HIV); (b) a small arrow indicating the entry route (e.g., an arrow from the mouth downward for cholera — 'contaminated water'; an arrow from the air toward the lungs for TB — 'airborne droplets'); (c) a brief label connecting the infection back to the phenomenon: for	Distribute or display the body-outline template. Say: 'Use three colours — green for fungal, red for bacterial, blue for viral. Show me WHERE in the body the infection acts and HOW the microorganism gets in. And for at least one fungal infection, write a phrase connecting it back to the ugali and chapati in Kisumu.' Set an 8-minute timer. Circulate and check that learners are drawing entry-route arrows, not just labelling body parts. Prompt where missing: 'How does Vibrio cholerae get from contaminated Lake Victoria water into someone's intestine? Show me that journey with an arrow.' After 8 minutes, invite 3 volunteers. For each, facilitate peer feedback: 'One star — something they did really well. One	Cumulative Annotated Model Building (NGSS Practice 2 — Developing and Using Models): by returning to and revising the same body-outline model across multiple lessons, learners build a coherent, progressively more sophisticated explanatory representation. The colour-coding makes the three microorganism categories perceptually distinct, reducing cognitive load while reinforcing classification. The entry-route arrows operationalise the mechanistic thinking required by the driving question — learners must trace the causal pathway from the microorganism's origin (e.g., the mould on food, contaminated lake water, airborne droplets) through the mode of transmission to the site of	Collect or photograph 5 randomly selected body-outline models. Check for: (1) correct colour-coding across all three categories; (2) at least one entry-route arrow per infection; (3) the fungal–phenomenon connection phrase. This provides a concrete, lesson-end snapshot of each learner's current explanatory model for use in planning Lesson 5 (Prevention and Control). Exit criterion: learners who have correctly placed and labelled at least 6 of the 8 infections (ringworm, oral thrush, cholera, typhoid, TB, influenza, HIV, COVID-19) with correct colours and at least one arrow are considered to have met the lesson SLO.

	ringworm, learners write 'same fungal kingdom as the mould on ugali and chapati.' This is a cumulative model — learners are told to use the same body outline in Lessons 5 and 6, where they will add prevention strategies and immune response details respectively. After 8 minutes, three volunteers briefly share their annotated diagrams under the document camera or by holding them up. The class gives one 'star' (something correct and well-labelled) and one 'wish' (something to improve) for each shared diagram.	wish — something they could add or improve.' Close by saying: 'Keep this diagram safe. You will add to it in every lesson until Lesson 8 — by then it will be your complete model of microorganisms and human health.'	infection, directly mirroring the explanatory structure demanded by the anchoring phenomenon.	
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your ARES teacher journal: (1) During the Jigsaw, which Expert Group had the most difficulty completing the four-column Evidence Table accurately — and what does that tell you about the quality of the Lesson 3 transmission-route instruction? (2) Did learners spontaneously use the word 'fungus' or 'fungi' when revisiting the ugali photograph, or did they still default to 'germs' or 'bacteria'? If the latter, plan a brief vocabulary reinforcement warm-up at the start of Lesson 5. (3) Were the structured-discussion anchor questions pitched at the right level — did all four generate genuine discussion, or did some fall flat? Note which questions to redesign for future cohorts. (4) Review the DQB new questions posted on orange sticky notes: do they reveal that learners are thinking mechanistically (how does the microorganism actually cause disease?) or are they still at a surface descriptive level (what is the name of the disease?)? The quality of these questions will determine how much scaffolding the Explain Phase 2 lesson (Lesson 5) needs to provide. (5) Note any learner who showed sensitivity or distress during the HIV/TB discussion and follow up privately — refer to the school counsellor if needed in line with your duty-of-care responsibility.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners examined ARES information cards showing photographs and descriptions of fungal infections (ringworm on a child's arm in Kisumu; oral thrush patches), bacterial infections (Vibrio cholerae linked to Lake Victoria water sources; Salmonella typhi; Mycobacterium tuberculosis in Nairobi informal settlements), and viral infections (influenza, HIV, COVID-19 in the Kenyan context). They gathered evidence through a Jigsaw activity, completing a four-column Evidence Table that recorded infection name, causative organism, mode of transmission, and body site affected.
What did I learn?	Learners learned that microorganisms causing human infections fall into three categories — fungi (same kingdom as the mould on the ugali and chapati), bacteria, and viruses — each with distinct causative organisms, transmission routes, and body sites affected. They learned that the mode of transmission identified in Lesson 3 (airborne, waterborne, direct contact, bodily fluids) directly predicts the type of infection that results, and that Kenyan communities near Lake Victoria face particular bacterial infection risks due to waterborne transmission of Vibrio cholerae.
How does this explain the phenomenon?	Learners constructed explanations linking the mould patches on the ugali and chapati to the broader world of microorganism-caused infections: the same fungal kingdom that spoils food overnight can infect human skin and mucous membranes; bacteria in

	contaminated water cause cholera and typhoid; viruses spread through droplets and bodily fluids to cause influenza, COVID-19, and HIV. These explanations were recorded in annotated body-outline models and three-sentence 'Explanation So Far' notebook entries, and were used to revise and extend the Driving Question Board.
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LESSON 5 (80 minutes): Prevention and Control: How Mode of Transmission Shapes Our Defences

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to explain how the mode of transmission of microorganisms determines appropriate prevention and control measures, and to evaluate the accessibility of those measures for Kenyan households.
Knowledge	Learners will be able to: (d) describe methods of control and prevention of infections caused by microorganisms, linking each method directly to the specific mode of transmission it interrupts.
Skills	Learners will design a community-facing prevention poster that accurately matches control strategies to transmission routes, and will evaluate which strategies are practical for low-income rural and urban Kenyan households.
Attitudes	Learners will appreciate the effect of microorganisms in day-to-day life and demonstrate respect for the lived realities of Kenyan families when proposing prevention strategies.
Key Inquiry Question	How are microorganisms important in day-to-day life?
Purpose in Storyline	Lessons 1–4 established what the fuzzy patches are (fungi, bacteria, some mould types), how they are classified, and how they are transmitted. Lesson 5 asks: now that we know HOW these organisms reach us, what can we ACTUALLY do to stop them — and who can afford to? Students move from describing the problem to designing community solutions, anchoring their analysis in the ugali-and-chapati phenomenon and in the real economic constraints of families in Kisumu, Kibera, and rural Rift Valley.
Safety Notes	No handling of visibly mouldy food samples. If photographs or sealed specimen jars are used, do not open jars in class. Wash hands before and after any practical activity. Remind learners that some mould spores are allergens — learners with known respiratory conditions should remain at a safe distance from any specimens. Dispose of any organic waste in a sealed bag.

B. LESSON OVERVIEW

Learners begin by revisiting the Driving Question Board entries from Lessons 1–4, then make a brief prediction about which prevention strategy would be MOST effective for the ugali phenomenon and WHY. The class views an annotated photographic case study of three Kisumu households — one urban low-income, one rural subsistence, one urban middle-income — each with different food-storage realities. Guided by a transmission-to-control matching activity, groups analyse how each mode of transmission (direct contact, airborne/droplet, contaminated food/water, vector-borne) maps onto a specific set of control measures. Each group then designs a bilingual (English/Kiswahili) prevention poster targeted at ONE infection category relevant to their assigned household type. Groups present, critique one another's posters using a structured peer-review protocol, and the teacher facilitates a whole-class equity analysis: which controls are accessible, which require resources many Kenyan families do not have, and what community-level alternatives exist. The lesson closes with a model-revision step and a DQB update that explicitly connects prevention logic back to the overnight spoilage of the ugali and chapati.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners individually write a 2-sentence prediction in their	Display the written prompt on the board or projector. SAY	Think-Pair-Share with Anchoring Phenomenon Re-entry. This	Observable indicator: learner written predictions name at least

 VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Control of Insects (Flexbook) Source: CK-12  Search ARES for similar readings	science notebooks in response to the prompt displayed on the board: 'We now know the fuzzy patches on the ugali and chapati are fungi and bacteria spread by airborne spores and contact. Predict: which ONE prevention strategy would have MOST reliably stopped the growth overnight — and explain why using evidence from Lessons 1–4.' After 3 minutes of silent writing, learners do a Think-Pair-Share with their seat partner. Selected pairs share with the class. Learners are then shown three photographs side by side: (A) ugali left uncovered on a counter in a Kisumu rental flat, (B) ugali covered with a wire mesh food dome (a common Kenyan kitchen item), and (C) ugali stored in a sealed plastic container in a fridge. They vote by show of hands on which scenario they predict will show least microbial growth after 24 hours, and briefly justify their vote to a neighbour.	VERBATIM: 'Before we learn anything new today, I want you to use only what you already know. Write your prediction — two sentences, your best scientific thinking, right now. Silence for three minutes.' Set a visible timer for 3 minutes. After writing, SAY: 'Turn to your partner — share your prediction and listen to theirs. You have 90 seconds.' After pairs discuss, cold-call FOUR pairs (spread across the room; aim for gender balance and at least one learner from a rural background). For each response, probe with: 'What evidence from our earlier lessons supports that prediction?' Do NOT affirm or correct yet — record key prediction words on the board under the heading PREDICTIONS. Then display the three photographs. SAY: 'Raise your hand for A... for B... for C. Now turn and tell your neighbour in one sentence WHY you chose that option.' WAIT 15 seconds before taking three volunteers. Capture the reasoning in a T-chart on the board: STRATEGY REASON. Leave predictions visible for the full lesson.	strategy activates prior knowledge from Lessons 1–4 and creates a public record of pre-instruction thinking that learners will revise by lesson end, making conceptual change visible and personally meaningful.	one specific transmission route (e.g., 'airborne spores') and link it to a prevention mechanism (e.g., 'covering stops spores settling'). Teacher scans notebooks during pair-share, noting learners who conflate transmission route with cause of spoilage — these learners will be targeted for guided questioning during the group activity.
Observe Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Control of Insects (Flexbook) Source: CK-12	Learners receive a printed or on-screen Kisumu Household Case Study Sheet. The sheet presents three household vignettes: (1) Mama Adhiambo's single-room rental in Nyalenda, Kisumu — no fridge, food stored on an open shelf, water fetched from a shared tap; (2) Mzee Otieno's farm homestead in Siaya County — earthen kitchen, grain stored in gunny sacks, meals cooked once daily; (3) The Kamau household in Milimani, Kisumu — fridge, running water, tiled kitchen surfaces. Each	Distribute materials and SAY VERBATIM: 'You are scientists and community health advisors. Your job is not just to know the science — it is to apply it fairly. Read the three household vignettes carefully. Then complete the matching table as a group. You have 12 minutes.' Circulate continuously. At the 4-minute mark, pause the class: SAY: 'I am hearing some groups link refrigeration to airborne transmission — is that the right match? Think about WHICH part	Case Study Analysis with Traffic-Light Equity Coding. By grounding the transmission-control mapping in real Kenyan household contexts, learners move beyond abstract science into applied epidemiology, while the traffic-light coding makes socioeconomic constraints a legitimate and explicit part of scientific reasoning rather than an afterthought.	Observable indicator: completed matching table rows correctly pair each transmission route with a mechanistically appropriate control method (e.g., covering food → interrupts airborne/droplet deposition of spores; hand-washing → interrupts direct contact route). Teacher checks that at least 3 of 4 rows are correctly completed before groups proceed to poster design. Equity coding accuracy is checked by teacher scan — a common error is marking ALL controls as RED for

Search ARES for similar readings	vignette lists the foods commonly stored overnight (ugali, sukuma wiki, leftover githeri, chapati, nyama choma) and common infections experienced by household members over the previous year (ringworm, cholera-like diarrhoea, typhoid, candidiasis, common colds). Alongside the vignettes is a Transmission-to-Control Matching Table with four rows (Direct Contact / Airborne & Droplet / Contaminated Food & Water / Vector-Borne) and four blank columns (Definition of Route, Example from the Phenomenon, Control Method, Why This Method Interrupts THIS Route). Learners work in groups of four to complete the table using the case study, their Lesson 1–4 notes, and a short reference card listing control vocabulary (refrigeration, boiling, hand-washing, covering food, vaccination, vector control, PPE, isolation, sterilisation). Groups annotate which controls are AVAILABLE to each household type using a traffic-light code: GREEN = readily available, AMBER = possible with effort or community support, RED = largely inaccessible.	of the transmission chain refrigeration actually interrupts. 10 seconds of silence.' Resume group work. At the 8-minute mark, prompt groups that are stuck on the Red/Amber/Green coding: SAY: 'Think about what Mama Adhiambo actually has in her kitchen right now. Is a fridge a green option for her? What might be a green option instead?' At 12 minutes, bring the class together. Cold-call THREE groups to share ONE row of their table each (rotate so all four rows are covered across the three groups). After each share, ask the class: 'Does anyone want to add or respectfully challenge that answer?' WAIT 10 seconds before taking responses. Annotate a class version of the table on the board.		low-income households; probe these groups with: 'What low-cost alternatives exist?'
Explain Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Control of Insects (Flexbook)	Each group is assigned ONE infection category from the matching table (fungal skin infections, bacterial foodborne illness, viral respiratory infections, or waterborne protozoal/bacterial infections) and ONE household type (Mama Adhiambo / Mzee Otieno / Kamau household). Groups design a bilingual (English and Kiswahili) A3 prevention poster targeted specifically at their assigned household. The poster	Assign groups by calling out names — do NOT allow self-selection to prevent ability clustering. Distribute A3 paper, markers, and the reference card. SAY VERBATIM: 'Your poster must speak directly to the family you have been assigned. Mama Adhiambo should be able to read it, understand it, and act on it with what she has TODAY. You have 12 minutes to design, then 90 seconds each to present.'	Poster Design as Argument Construction + Structured Equity Discussion. Poster design requires learners to synthesise transmission mechanism, control logic, and audience context into a single coherent argument — a high-order application of NGSS Science and Engineering Practice 6 (Constructing Explanations) and Practice 8 (Obtaining, Evaluating, and Communicating Information). The equity discussion enacts KICD	Observable indicator: poster includes a mechanistically correct transmission-to-control link (not just a list of hygiene tips) AND contains a visible connection to the ugali/chapati phenomenon. Teacher uses a rapid 3-point rubric during presentations: 3 = correct mechanism + equity ranking + phenomenon link; 2 = correct mechanism + phenomenon link but no equity ranking; 1 = list of controls with no mechanistic

Source: CK-12 Search ARES for similar readings	must include: (a) the name and brief description of the infection category; (b) the specific mode of transmission; (c) at least THREE control/prevention measures ranked by accessibility (most accessible first); (d) one direct visual or caption connection to the ugali-and-chapati phenomenon; (e) a 'Did You Know?' fact box citing a Kenyan health statistic or referencing the Kenya Ministry of Health guidelines. Groups present their poster to the class in a 90-second pitch. After all presentations, the class participates in a structured whole-class equity analysis discussion led by the teacher: Which control measures appeared on EVERY poster (universal controls)? Which appeared only on the Kamau household poster (resource-dependent)? What community-level strategies — e.g., women's group food safety sensitisation in Siaya, school hygiene clubs in Nyalenda — could bridge the gap? The teacher draws a two-column summary on the board: INDIVIDUAL-LEVEL CONTROLS COMMUNITY/SYSTEMS-LEVEL CONTROLS.	Circulate during design time. At the 6-minute mark, prompt groups not yet including the ugali connection: SAY: 'I do not yet see the ugali and chapati on your poster. How does YOUR infection category connect to what happened in that Kisumu kitchen overnight?' During presentations, enforce the 90-second limit with a visible timer. After each presentation, invite one peer question using the sentence stem displayed on the board: 'I notice your poster says ____ — can you explain how that specifically interrupts ____?' For the equity analysis discussion, SAY: 'Let us be honest scientists AND honest citizens. Put your hand up if you think Mama Adhiambo could realistically carry out EVERY control on the board right now.' WAIT 15 seconds. Then: 'What does science owe communities that cannot afford the textbook answer? What are the alternatives?' Cold-call FIVE learners across the room, including at least two who have been quiet. Capture responses in the two-column summary.	values of social justice and responsibility.	explanation. Groups scoring 1 are flagged for targeted support in Lesson 6.
Driving Question Board (DQB) Creation  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners return to the class Driving Question Board (DQB) established in Lesson 1. Each learner receives two sticky notes of different colours: ORANGE = 'A question Lesson 5 has answered for me' and BLUE = 'A new question Lesson 5 has opened up'. Learners write one entry on each colour and place them on the DQB in the appropriate column. The class does a 3-minute silent gallery walk of the new additions. Then three volunteers read aloud	Distribute sticky notes and SAY VERBATIM: 'Orange note: write one specific question that Lesson 5 has now answered for you — be precise, not vague. Blue note: write one new question this lesson has opened up. You have 3 minutes, working silently.' After placement on the DQB, SAY: 'Silent gallery walk — read what your classmates wrote. You have 3 minutes.' After the walk, cold-call three learners to read one of each colour. For each blue (new	Driving Question Board as Metacognitive Tracker. The two-colour system makes the distinction between resolved and emerging understanding visible to both learners and teacher, supporting NGSS Practice 1 (Asking Questions) and providing real-time formative data on conceptual gaps that should inform Lesson 6 planning.	Observable indicator: orange notes name a specific scientific concept (e.g., 'I now understand why covering food interrupts airborne transmission of mould spores') rather than a vague statement ('I learned about germs'). Teacher collects all blue notes at the end of class, sorts them by theme overnight, and uses the most frequent themes to adjust the opening hook of Lesson 6.

 HTML: Control of Insects (Flexbook) Source: CK-12  Search ARES for similar readings	one orange and one blue note each. The teacher uses the blue (new question) notes to preview Lesson 6: 'Several of you are now asking about the ECONOMIC importance of microorganisms — not just the harmful side. That is exactly where we go next.' The teacher also explicitly connects Lesson 5 back to the driving question by SAYing VERBATIM: 'We started this unit because ugali and chapati grew fuzzy patches overnight in a Kisumu kitchen. Today you designed science-based, equity-aware advice that family could actually use. That is applied microbiology. Write in your notebook: covering the sufuria and storing food away from warm surfaces interrupts the airborne and contact transmission routes that allowed the mould spores to settle and germinate overnight.'	question) note, ask the class: 'Can anyone partially answer this right now — or do we need more evidence?' WAIT 10 seconds. Use the blue notes to draw an explicit arrow on the DQB toward the Lesson 6 column. End with the verbal summary statement connecting to the phenomenon, written by learners in their notebooks.		
Model Building Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Control of Insects (Flexbook) Source: CK-12  Search ARES for similar readings	Learners open their cumulative model of the ugali-and-chapati phenomenon, first drawn in Lesson 1 and revised in each subsequent lesson. Using a coloured pen different from all previous revisions, each learner adds a PREVENTION LAYER to their model. This layer must show: (a) at least two physical barriers or interventions drawn onto the diagram (e.g., a lid on the sufuria, a refrigerator symbol, a hand-washing icon near the food); (b) an arrow labelled 'INTERRUPTED' crossing out or blocking one of the transmission pathways already drawn in a previous lesson; (c) a note in the margin: 'Accessible to low-income household? YES / PARTIALLY / NO — because ____.' Learners share their revised model with one neighbour and give one	SAY VERBATIM: 'Take out your cumulative model. Pick up a colour you have NOT used in any previous lesson — this is your prevention layer. You have 7 minutes to revise your model. Your model should now tell the FULL story: what the organisms are, how they got onto the ugali and chapati, AND what could have stopped them.' Set a timer. Circulate and prompt learners who are only adding labels without showing mechanism: SAY: 'Drawing a fridge is good — now draw an arrow showing WHICH part of the transmission chain the fridge interrupts and label it.' At 7 minutes, SAY: 'Share with your neighbour. Use the sentence stem on the board. 90 seconds.' After sharing, collect models. SAY: 'Every time you revise this model,	Cumulative Model Revision as Epistemic Record. The multi-colour layering system makes the progression of scientific understanding physically visible across the unit, enacting NGSS Practice 2 (Developing and Using Models) and giving learners ownership of their growing explanatory framework. The equity annotation integrates KICD values of social justice directly into the scientific model.	Observable indicator: the prevention layer correctly places an intervention at the biologically appropriate point in the transmission chain (e.g., a lid drawn at the point where airborne spores descend onto the food surface, not randomly placed). Teacher checks during collection that ALL three required components (barrier, interrupted arrow, equity note) are present. Models missing the interrupted arrow are flagged — these learners have not yet connected prevention mechanism to transmission route and need re-teaching at the start of Lesson 6.

	piece of feedback using the sentence stem: 'I can see you have shown ____ — have you also considered ____?' Models are collected by the teacher for portfolio assessment.	your understanding of the phenomenon grows. By Lesson 8, this model will explain everything — from the invisible spore to the community health response.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. **EQUITY ANALYSIS DEPTH:** Did learners move beyond listing controls to genuinely interrogating WHY some measures are inaccessible? If the equity discussion stayed superficial (e.g., learners said 'buy a fridge' without questioning affordability), consider opening Lesson 6 with a brief roleplay where one learner plays Mama Adhiambo responding to advice she cannot afford.
2. **POSTER MECHANISM ACCURACY:** How many groups correctly linked their control measure to the specific point in the transmission chain it interrupts, versus groups that listed hygiene tips generically? Review the 3-point rubric scores. If more than two groups scored a 1, the transmission-to-control mapping concept needs reinforcement in Lesson 6 before moving to economic importance.
3. **DQB BLUE NOTES:** Review the blue (new question) sticky notes collected at the end of class. Tally the themes. If a significant proportion of questions concern the BENEFICIAL uses of microorganisms, you have strong intrinsic motivation to anchor the economic importance lesson (Lesson 6) in learner curiosity rather than curriculum sequencing alone.
4. **PHENOMENON CONNECTION:** Did every group's poster reference the ugali-and-chapati phenomenon explicitly, or did some drift into generic health education? The phenomenon must remain the anchor throughout — if posters lost the connection, remind learners at the start of Lesson 6 that ALL their science this unit exists to explain that one overnight event in Kisumu.
5. **DIFFERENTIATION NOTE:** Learners who are from low-income backgrounds may have found the equity discussion personally sensitive. Check in privately with any learner who appeared disengaged or distressed. The goal is empowerment through science, not exposure of personal circumstances.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Three Kisumu household vignettes (Mama Adhiambo in Nyalenda, Mzee Otieno in Siaya, the Kamau household in Milimani) showing different food-storage realities, infection histories, and available resources; photographs of covered versus uncovered versus refrigerated ugali; a Transmission-to-Control Matching Table completed in groups; peer prevention posters.
What did I learn?	The mode of transmission of a microorganism determines which control measure is most effective: covering food interrupts airborne/droplet deposition of spores (fungal and bacterial); hand-washing interrupts direct contact transmission; boiling water and cooking food thoroughly interrupts the contaminated food/water route; vector control and sanitation interrupt vector-borne and faecal-oral routes. Not all scientifically valid controls are equally accessible — low-income households in rural Kenya (e.g., Siaya County) and urban informal settlements (e.g., Nyalenda, Kisumu) may lack refrigeration, running water, or sealed storage, requiring community- and systems-level alternatives such as women's group sensitisation, school hygiene clubs, and low-cost food-covering practices.
How does this explain the phenomenon?	The overnight growth of white, green, and black fuzzy patches on the ugali and chapati in the Kisumu kitchen occurred because the uncovered food surface provided an unobstructed landing site for airborne fungal spores (and bacterial cells) already present in the kitchen environment. Covering the sufuria — even with a simple wire mesh dome or a clean cloth, which costs nothing — would have physically blocked the airborne transmission route and dramatically reduced microbial colonisation. Refrigeration would additionally have slowed the germination and growth of any spores that did land, by reducing the temperature below the optimum

	growth range of most mesophilic fungi and bacteria. The variation in patch colour (white, green, black) across the same food surface reflects colonisation by DIFFERENT species of mould (e.g., Rhizopus, Penicillium, Aspergillus), each arriving via the same airborne route but occupying slightly different microhabitats on the food substrate.
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LESSON 6 (80 minutes): Economic Importance of Microorganisms: The Dual Nature of the Invisible World

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To explain the economic importance of microorganisms — both their beneficial roles in food production, agriculture, medicine, and energy, and their harmful roles in food spoilage and crop disease — and to connect this dual nature back to the ugali and chapati phenomenon.
Knowledge	Learners will be able to explain the economic importance of microorganisms, including: beneficial roles (fermentation in yoghurt/uji by <i>Lactobacillus</i> , leavening of bread/mandazi by <i>Saccharomyces</i> /yeast, nitrogen fixation by <i>Rhizobium</i> in Kenyan farmland, antibiotic production from <i>Penicillium</i> , biogas generation from methane-producing bacteria, decomposition for soil fertility) and harmful roles (food spoilage causing post-harvest losses, crop diseases such as maize rust and coffee berry disease).
Skills	Learners will be able to: (1) classify microorganisms as economically beneficial or harmful with evidence; (2) construct a dual-column concept organiser linking microorganism names to specific economic roles; (3) participate in structured peer-sharing to synthesise multiple lines of evidence; (4) update the Driving Question Board with new evidence statements.
Attitudes	Learners will appreciate the dual nature of microorganisms — recognising that the same invisible organisms responsible for spoiling the ugali and chapati in Kisumu are close relatives of organisms that sustain Kenyan livelihoods through yoghurt production, bread-making, soil fertility, and medicine.
Key Inquiry Question	How are microorganisms important in day-to-day life?
Purpose in Storyline	Lessons 1–5 established what microorganisms are, how they are classified (fungi, bacteria, viruses), how they are transmitted, and how they cause infections. Lesson 6 now pivots from harm to the full economic picture: the same group of organisms that ruined the Kisumu family's ugali overnight also makes their uji ferment, their bread rise, their soil fertile, and their antibiotics possible. This dual-nature reveal is the conceptual turning point of the unit — students must revise their mental model of microorganisms from 'agents of disease and spoilage' to 'double-edged biological agents whose impacts depend on context, species, and conditions.'
Safety Notes	No live cultures or food preparation required in this lesson. If food samples (commercial yoghurt, mandazi, bread) are brought in as artefacts for discussion, they must be commercially packaged and unopened. No tasting of any food items in the classroom. Learners with known mould or food allergies should not handle opened food packages. Wash hands before and after any handling of food packaging. All waste to be disposed of in the classroom bin.

B. LESSON OVERVIEW

In this 80-minute Explain Phase lesson, learners build on four prior lessons of evidence-gathering (microorganism types, transmission, infections, and control) to now investigate the full economic importance of microorganisms. The lesson opens by reconnecting to the anchoring phenomenon — the fuzzy patches on the ugali and chapati in Kisumu — and asking: if these organisms can destroy food overnight, can related organisms also create economic value? Through a structured Think-Pair-Share, a jigsaw expert-group activity, and a whole-class gallery walk, learners analyse six beneficial economic roles (fermentation, leavening, nitrogen fixation, antibiotic production, biogas generation, and decomposition) and two harmful economic roles (food spoilage losses, crop diseases). Learners construct a dual-column Benefit–Harm concept organiser, present findings to peers, and update the Driving Question Board. The lesson closes with a 3-2-1 exit ticket that directly references the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Importance of the Oceans (Flexbook) Source: CK-12  Search ARES for similar readings	Learners are shown the same projected photograph of the spoiled ugali and chapati from Lesson 1 — white, green, and black fuzzy patches visible on both foods. Teacher then projects a second image alongside it: a vibrant Kisumu market stall displaying sealed cups of commercial yoghurt, a tray of freshly baked mandazi, and a bag of Unga maize flour with a 'fortified' label. Learners are given 2 minutes of silent individual writing time to respond to the prompt written on the board: 'The organisms responsible for the fuzzy patches on the ugali belong to the same broad group as organisms used to make yoghurt, mandazi, and to fertilise Kenyan farmland. Do you believe this statement is true or false? Write three reasons for your position.' After writing, learners share predictions with a shoulder partner for 2 minutes. Selected pairs share with the class.	Display both images side-by-side on the projector and say: 'Look carefully at both images. On the left — the spoiled ugali and chapati from our Kisumu family. On the right — products made with the help of microorganisms sold right here in Kenyan markets. I want you to think silently for 2 minutes and write your answer to this question on your jotter.' [Point to the board prompt.] After 2 minutes, say: 'Turn to your shoulder partner and share what you wrote. You have 90 seconds.' [Set a visible timer.] After partner share, cold-call 3 pairs: 'Pair at the front-left window — what did you predict and why?' [WAIT 10 seconds after each response before probing.] After 3 pairs, say: 'Interesting — we have some who say true and some who say false. By the end of today's lesson, you will have enough evidence to settle this question permanently and update our Driving Question Board.'	Prediction with justification (Predict–Justify): By requiring three written reasons before discussion, this strategy activates prior knowledge from Lessons 1–5, surfaces misconceptions (e.g., 'beneficial organisms cannot be related to harmful ones'), and creates cognitive dissonance that motivates the evidence-gathering phases that follow. The side-by-side image juxtaposition anchors the economic lens directly to the phenomenon.	Teacher circulates during silent writing and reads over shoulders, noting: (1) whether learners reference specific microorganism names from previous lessons (fungi, bacteria); (2) whether any learners spontaneously mention fermentation or decomposition (indicating prior knowledge); (3) whether misconceptions about spontaneous generation resurface. Teacher mentally flags 2–3 learners with strong predictions and 2–3 with misconceptions for targeted cold-calling in later phases.
Observe Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Importance of the Oceans (Flexbook)	Learners are divided into 6 expert groups of approximately 5–6 learners each. Each group receives a laminated A3 Resource Card describing one specific economic role of microorganisms, written in accessible English with Kenyan contexts. The six topics are: (A) Fermentation — Lactobacillus in yoghurt and fermented uji; (B) Leavening — Saccharomyces cerevisiae in bread and mandazi; (C) Nitrogen Fixation — Rhizobium in bean and maize farms in the Rift Valley; (D)	Before distributing cards, say: 'Each group is now an expert team. Your job is to read your card carefully, understand the economic role described, and post evidence on our Benefit–Harm wall chart. You will then teach the rest of the class what you found.' Distribute cards and say: 'You have 15 minutes. Every member of your group must be able to explain the key idea — not just the best reader.' [Set a 15-minute visible timer.] Circulate to each group during reading, crouching to eye	Jigsaw with Gallery Walk: The jigsaw structure ensures every learner develops deep familiarity with one economic role before encountering the others, distributing cognitive load across the class. The Gallery Walk then re-exposes all learners to all six roles through peer-generated notes rather than teacher transmission, building a shared evidentiary base. The question sticky notes (red) make visible the class's remaining uncertainties — these will feed directly into the	Observable indicators during group work: (1) All group members can name the microorganism and at least one Kenyan context without referring to the card — teacher checks this by asking random non-readers in each group. (2) Sticky notes on the wall chart are specific (e.g., 'Lactobacillus converts milk sugars to lactic acid, making yoghurt') rather than vague ('bacteria are useful'). (3) During Gallery Walk, red question sticky notes are substantive (e.g., 'How does

Source: CK-12 Search ARES for similar readings	Antibiotic Production — Penicillium mould and the discovery of penicillin; (E) Biogas Generation — methane-producing bacteria and biogas digesters used by Kenyan rural households; (F) Decomposition and Soil Fertility — bacteria and fungi breaking down dead organic matter to return nutrients to Kenyan farm soil. Each group also receives a sticky-note pack and a section of the classroom Benefit–Harm wall chart (a large sheet of paper divided into two columns). Groups spend 15 minutes reading their resource card together, discussing it, and writing key facts on sticky notes to post on the wall chart. In the final 15 minutes, groups rotate in a Gallery Walk — each group visits every other group's wall-chart section, reads the sticky notes, and adds a 'question' sticky note (a different colour) if they have a query about that section.	level and asking: 'What is the name of the microorganism on your card?' [WAIT 10 seconds.] 'Where in Kenya would you see this happening?' [WAIT 12 seconds.] For Group C (nitrogen fixation), prompt if needed: 'Think about bean farming in Kakamega or the Rift Valley — what happens to the soil after beans are grown?' After 15 minutes, announce the Gallery Walk: 'Now you will spend 2 minutes at each other group's section. Read their sticky notes. If something surprises you or you have a question, write it on a red sticky note and add it to their section.' [Rotate groups every 2 minutes using a timer.] After the Gallery Walk, cold-call one learner from each group to read ONE surprising thing they learned from another group's section. [WAIT 10 seconds per response.]	DQB update in Phase 4.	Rhizobium actually put nitrogen into the soil?') rather than off-topic. Teacher collects and reads all red sticky notes after the Gallery Walk and uses them to prioritise content in Phase 3.
Explain Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Importance of the Oceans (Flexbook) Source: CK-12 Search ARES for similar readings	Teacher leads a whole-class structured discussion to synthesise the evidence from the Gallery Walk, explicitly connecting each economic role back to the anchoring phenomenon. Teacher projects a blank Dual-Nature Concept Organiser (a T-chart with columns 'Beneficial Economic Role' and 'Harmful Economic Role') on the board and builds it live with learner contributions. After completing the Beneficial column from the six expert groups, teacher introduces the Harmful column: learners are guided through two harmful economic roles — (1) food spoilage and post-harvest losses (linking directly to the Kisumu ugali/chapati phenomenon, and extending to	Open Phase 3 by saying: 'Let us now bring all the evidence together and answer the big question — are microorganisms economically good, bad, or both?' Draw the T-chart live on the board. Point to the wall chart and say: 'I will call on each expert group to give me their ONE most important economic benefit. Listen carefully because you will need this for your notebook.' Cold-call one spokesperson from each of the six groups in sequence. [WAIT 12 seconds after each before writing on the board.] After the Beneficial column is complete, say: 'Now let us look at the Harmful side — and here is where we return to the Kisumu family.' Project the spoiled ugali image again. Ask: 'Based on	Live Concept Organiser Construction (Co-constructed T-chart): By building the organiser in real time with learner contributions rather than revealing a completed version, this strategy makes thinking visible, allows the teacher to correct misconceptions immediately, and gives learners ownership of the knowledge structure. The 'Explain It to a Market Vendor' paragraph is a disciplinary literacy move — translating scientific knowledge into a real-world communication task grounded in the Kenyan economic context of the phenomenon.	Teacher checks notebooks of at least 6 learners during the 5-minute writing time, specifically looking for: (1) correct use of at least two microorganism names (e.g., Lactobacillus, Saccharomyces, Penicillium, Rhizobium); (2) accurate distinction between beneficial and harmful roles; (3) a direct connection to the phenomenon (ugali/chapati spoilage or food vendor losses). Learners who cannot name two microorganisms are identified for additional support. The three cold-called readings provide whole-class formative data — teacher notes whether the class can articulate the dual-nature concept clearly.

	market vendors losing income from spoiled githeri and sukuma wiki); and (2) crop diseases such as maize rust caused by fungal pathogens and coffee berry disease affecting Kenyan highland farms in Kericho and Nyeri. Learners copy the completed organiser into their science notebooks. The class then engages in a Socratic discussion: 'Is the mould on the ugali in our photograph economically harmful? Could it ever be economically beneficial? What would need to change?' Learners write a 3-sentence 'Explain It to a Market Vendor' paragraph in their notebooks — imagining they are explaining to a Kisumu market vendor why the same organisms that spoil her mandazi are related to those that make her yoghurt sell.	what we know about fungi and bacteria, what economic damage is the mould on this ugali causing — not just to this family, but to vendors across Kisumu market?' [WAIT 15 seconds. Cold-call 2 learners.] Then say: 'In Kenya, post-harvest food losses cost farmers and vendors billions of shillings annually. This is the same biology we are studying.' Introduce crop diseases: 'Has anyone seen rusty-orange patches on maize stalks in a shamba near your home? That is a fungal crop disease — same kingdom as the mould on our ugali.' After completing the organiser, say: 'Now, in your notebooks, write 3 sentences explaining the dual nature of microorganisms to a Kisumu mandazi vendor. Use the names of at least two microorganisms.' [Allow 5 minutes of silent writing.] Cold-call 3 learners to read their sentences. [WAIT 10 seconds before selecting next reader.]		
Driving Question Board (DQB) Creation  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Importance of the Oceans (Flexbook) Source: CK-12  Search ARES	Learners return to the class Driving Question Board — a large display maintained since Lesson 1 that tracks the class's growing understanding of the phenomenon. Each learner receives two sticky notes: one green ('I now know...') and one orange ('I still wonder...'). Learners silently write one statement on each sticky note in response to the driving question: 'Why did the ugali and chapati grow fuzzy patches overnight — and how can understanding the invisible organisms responsible help us protect our health and our food in Kenya?' Learners post their sticky notes on the DQB. Teacher reads a selection aloud and	Say: 'It is time to update our Driving Question Board. Think back to where we started in Lesson 1 — we were puzzled by fuzzy patches on ugali. Look at how much evidence we have gathered.' [Gesture to the full DQB.] 'Take your two sticky notes. On the GREEN one, write one thing you now KNOW that helps answer our driving question. Be specific — use a microorganism name or an economic role we discussed today. On the ORANGE one, write one thing you still WONDER about.' [Allow 3 minutes of silent writing. Set a timer.] 'Now come up and post them in the correct column.' [Give learners 1	Driving Question Board Update (Metacognitive Anchoring): The DQB is a long-term sensemaking artefact that externalises the class's epistemic progress. By requiring learners to distinguish what they know from what they wonder, this strategy builds metacognitive awareness and maintains the driving question as the coherent thread across all 8 lessons. The teacher's act of labelling orange notes with 'L7' or 'L8' signals to learners that their questions are legitimate and will be answered, sustaining intellectual engagement.	Teacher photographs the DQB after sticky notes are posted for later analysis. Observable indicators: (1) Green notes should reference specific microorganism names, economic roles, or the phenomenon — vague notes ('microorganisms are useful') indicate insufficient depth and those learners should be targeted in Lesson 7. (2) Orange notes reveal remaining misconceptions (e.g., 'Can you eat the mould on ugali if it is the same as the mould that makes penicillin?') — teacher notes these for explicit address in upcoming lessons. (3) The ratio of green to orange notes gives a class-level snapshot of conceptual

for similar readings	facilitates a 4-minute discussion to identify what the class now collectively understands and what questions remain open for Lessons 7 and 8.	minute to post.] Read 3 green notes aloud, pausing after each: 'This learner writes — [read note]. Does the class agree this answers part of our driving question? Thumbs up or thumbs sideways.' Then read 2 orange notes: 'This is a question we will address in Lesson 7 — prevention and control — and Lesson 8 — our final model.' Circle orange notes that relate to prevention/control and write 'L7' next to them, and those relating to model-building write 'L8'.		progress at the lesson midpoint of the unit.
Model Building Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Importance of the Oceans (Flexbook) Source: CK-12  Search ARES for similar readings	Learners open their science notebooks to their personal model of microorganism activity — a diagram or annotated drawing they have been revising since Lesson 1. They are given 4 minutes to add new features to their model reflecting today's learning: specifically, they must add at least two new annotations showing beneficial economic roles and one annotation showing a harmful economic role, using arrows or labels. Learners who finish early add a 'context label' showing where in Kenya each role occurs (e.g., 'Rhizobium → bean shamba, Rift Valley'). In the final 2 minutes, teacher projects one volunteer learner's model (with permission) using a document camera or phone camera and the class briefly discusses what the revised model now explains about the driving question that the original model from Lesson 1 could not.	Say: 'Open your notebooks to your microorganism model. This model has been growing since Lesson 1. Today we have learned that the same invisible world that caused the fuzzy patches on the Kisumu family's ugali is also responsible for making yoghurt, building soil fertility on Kenyan farms, and producing the antibiotics that save lives. Your model must now show this dual nature.' [Write on the board: 'Add to your model: (1) Two beneficial economic roles with the microorganism name and a Kenyan location. (2) One harmful economic role linked back to the ugali/chapati phenomenon.'] 'You have 4 minutes. Make your annotations as specific as possible — I should be able to read your model and understand both sides of the story.' [Circulate and observe. After 3 minutes, quietly ask one learner with a strong model: 'May I show your model to the class for 1 minute at the end?'] After 4 minutes, project the selected model. Say: 'Look at this model. What does it now show that our very first model in Lesson 1 did not show?' [WAIT 12 seconds.	Cumulative Model Revision: Returning to and annotating the same model across all 8 lessons makes the growth of understanding visible and tangible. Each revision forces learners to integrate new knowledge with prior knowledge rather than treating each lesson as isolated content. The public projection of one model activates social learning — peers evaluate and critique the model collectively, sharpening the whole class's understanding of what a complete explanatory model must include.	During the 4-minute model revision, teacher checks a minimum of 6 notebooks against three criteria: (1) At least two named beneficial microorganisms are present with their roles (e.g., 'Saccharomyces → leavening → mandazi rises'); (2) A harmful role is explicitly linked to the ugali/chapati phenomenon with an arrow or annotation; (3) At least one Kenyan geographic context label is present. Learners missing two or more criteria receive a written prompt in their notebook margin: 'Add: [specific missing element] — see today's T-chart.' The teacher's observation notes from this phase directly inform the differentiation plan for Lesson 7.

		Cold-call 2 learners.] Close by saying: 'In Lesson 7, we will use everything we now know — about what microorganisms are, how they spread, and how important they are economically — to answer the final part of our driving question: how can we actually protect our health and our food in Kenya?'	
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record responses in your professional journal:

1. DUAL-NATURE CONCEPTUAL SHIFT: Did the majority of learners successfully articulate the dual nature of microorganisms in their 'Explain It to a Market Vendor' paragraph? Could they name specific organisms (Lactobacillus, Saccharomyces, Rhizobium, Penicillium) linked to specific roles, or did responses remain vague? What does this tell you about the depth of learning across Lessons 1–5?
2. JIGSAW EQUITY: In the expert groups, were all learners engaged, or did stronger readers dominate? Was the 'every member must explain' instruction sufficient to distribute participation? If not, what structured role cards (e.g., Reader, Recorder, Reporter, Timekeeper, Questioner) should be introduced for Lesson 7?
3. DQB ANALYSIS: Review your photograph of the DQB. Are the green sticky notes substantively linked to the phenomenon and to specific microorganism names? Are the orange 'wonder' notes pointing toward prevention/control (Lesson 7) or toward deeper conceptual questions about microbial life? Do any orange notes reveal persistent misconceptions that must be explicitly addressed?
4. KENYAN CONTEXT RESONANCE: Which Kenyan context generated the most engagement — yoghurt/uji, mandazi leavening, biogas digesters, Rift Valley nitrogen fixation, or crop diseases? Use this to prioritise contexts in Lesson 7 and Lesson 8.
5. LESSON 7 READINESS: Based on exit ticket data and notebook checks, identify by name any learners who cannot yet distinguish beneficial from harmful microorganism roles. Plan a 5-minute targeted review at the start of Lesson 7 for this group before introducing prevention and control methods.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a juxtaposition of the spoiled ugali/chapati phenomenon photograph alongside images of Kenyan commercial yoghurt, mandazi, and maize flour — products linked to microorganism activity. They also observed peer-generated sticky notes on a Benefit–Harm wall chart produced during the jigsaw expert group activity, and a classmate's annotated cumulative model projected for whole-class analysis.
What did I learn?	Learners learned that microorganisms have a dual economic nature: (1) Beneficial roles — Lactobacillus ferments milk/uji, Saccharomyces leavens bread and mandazi, Rhizobium fixes atmospheric nitrogen in Kenyan bean and maize farms, Penicillium produces penicillin antibiotics, methane-producing bacteria generate biogas for rural Kenyan households, and decomposer bacteria/fungi recycle nutrients to maintain soil fertility; (2) Harmful roles — fungal and bacterial spoilage of food (directly causing the ugali/chapati phenomenon) generates significant post-harvest economic losses for Kenyan market vendors, and fungal pathogens cause crop diseases such as maize rust and coffee berry disease. The same biological kingdom (e.g., fungi) contains both spoilage agents and commercially vital organisms.
How does this explain the phenomenon?	Learners explained, through a co-constructed T-chart concept organiser, a written 'Explain It to a Market Vendor' paragraph, and an updated cumulative model, that the microorganisms responsible for the fuzzy patches on the Kisumu family's ugali and chapati

	belong to the same broad taxonomic groups as organisms that underpin critical sectors of Kenya's food economy, agricultural productivity, and healthcare system. The context-dependence of microorganism impact — harmful when growing unchecked on stored food, beneficial when harnessed in controlled fermentation, farming, or pharmaceutical processes — was articulated as the central explanatory principle. Learners updated the Driving Question Board to reflect this dual-nature understanding and identified open questions about prevention and control to be addressed in Lesson 7.
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LESSON 7 (40 minutes): Project Investigation: Mould Growth on Food Substrates — What Conditions Do Microorganisms Need to Thrive?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students will conduct and analyse a hands-on investigation into the conditions that favour mould growth on common Kenyan food substrates, generating first-hand evidence to explain why the ugali and chapati in the phenomenon spoiled overnight and to establish the scientific basis for food preservation strategies.
Knowledge	Students will be able to identify the environmental conditions (moisture, warmth, air exposure) that promote or inhibit microbial growth on food substrates; explain how these conditions relate to the spoilage observed in the anchoring phenomenon; and connect their findings to the food preservation principles discussed in Lesson 5.
Skills	Students will set up a fair investigation by identifying independent, dependent and controlled variables; make systematic observations using their senses and photographs; record quantitative and qualitative data in a structured table; and draw evidence-based conclusions that link experimental results to real-world food preservation decisions in Kenyan households.
Attitudes	Students will demonstrate curiosity and patience in scientific observation, intellectual honesty in recording results even when they differ from predictions, and a sense of responsibility in connecting laboratory findings to community health and food security in Kenya.
Key Inquiry Question	How are microorganisms important in day-to-day life? — specifically, what conditions allow microorganisms to grow on food, and how can understanding those conditions help Kenyan families protect their food?
Purpose in Storyline	Lesson 7 is the Observe and Analyse phase of the evidence-gathering sequence. After six lessons of building conceptual knowledge about microorganism types, transmission, infections, prevention and economic importance, students now generate their own empirical evidence about microbial growth conditions. This first-hand data advances the driving question by providing concrete experimental proof for why the uncovered, warm, moist ugali and chapati in Kisumu were ideal substrates for overnight mould growth, and sets up the model revision and DQB closure in Lesson 8.
Safety Notes	Do not open mould containers and inhale directly — mould spores can irritate airways. Students with asthma or known respiratory sensitivities should observe pre-sealed transparent containers only and must not handle open samples. Wash hands thoroughly with soap and water after handling any food samples. Dispose of all mouldy samples in sealed plastic bags in the bin — do not compost. Teacher should prepare pre-incubated samples in sealed clear bags or containers so that visual observation is possible without opening. No eating of any food samples used in the investigation.

B. LESSON OVERVIEW

In this 40-minute investigation lesson, students first predict which conditions will produce the most mould growth on three food substrates (ugali, bread, dry maize biscuit). They then observe and analyse both freshly set-up samples and teacher-prepared pre-incubated samples that show visible mould growth under different conditions (covered or uncovered, moist or dry, warm or cool). Students photograph and record their observations in a structured data table, identify patterns, and draw conclusions about the environmental requirements of moulds. The lesson explicitly reconnects every finding to the anchoring phenomenon — the overnight spoilage in Kisumu — and adds new empirical evidence to the Driving Question Board, preparing students for the model revision and community message tasks in Lesson 8.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Exponential Growth and Decay (Flexbook) Source: CK-12  Search ARES for similar readings	Students are reminded of the anchoring phenomenon by viewing the photograph of the spoiled ugali and chapati on the board. The teacher briefly recaps what students established in Lesson 6 — that fungi cause food spoilage and represent economic loss. The teacher then presents the investigation question written on the board: Which conditions cause mould to grow fastest on ugali, bread and a dry maize biscuit? Students examine a labelled diagram on the board showing eight sample containers set up across four conditions: Container A — ugali, uncovered, moist, warm; Container B — ugali, covered, moist, cool; Container C — bread, uncovered, moist, warm; Container D — bread, covered, dry, cool; Container E — dry maize biscuit, uncovered, warm; Container F — dry maize biscuit, covered, cool; plus two teacher pre-incubated versions of A and C labelled Pre-A and Pre-C for immediate observation. In their notebooks students write their predictions for each container using the sentence frame: I predict Container ____ will show the most mould because _____. They also identify the independent variable (the condition being changed), the dependent variable (amount and type of mould growth) and state two controlled variables (same amount of food, same room, same time period). Pairs share predictions with a neighbouring pair before sharing with the class. Teacher records a class prediction	Teacher displays the phenomenon photograph and says: Cast your mind back to Kisumu — uncovered ugali, overnight, white green and black patches. Today we stop wondering and start testing. Our investigation asks: what exactly does a mould need to grow? Before we look at any results, I want your prediction. Use the sentence frame in your notebook. You have two minutes — go. [WAIT TIME: 2 minutes silent individual writing.] Teacher circulates, noting predictions. After 2 minutes: Now share your prediction with your partner. Justify it — do not just say warm, say WHY warm. [WAIT TIME: 1 minute pair talk.] Teacher cold-calls three pairs and records tallies on the board. Teacher then asks: What is the one thing we are changing between Container A and Container B? [WAIT TIME: 10 seconds.] After student response: Good. That one thing we deliberately change is called the independent variable. What are we measuring or observing to see if our change made a difference? [WAIT TIME: 10 seconds.] Teacher confirms dependent variable. Teacher then asks: If we want our test to be fair, what must stay the same across all containers? [WAIT TIME: 15 seconds, take two responses.] Teacher writes the three variable types on the board and says: This is what makes our investigation scientific rather than just guessing — it is the same logic scientists at KEMRI use when they test how	Prediction with justification using sentence frames activates prior knowledge from Lessons 2, 3 and 5 about mould structure, spore transmission and conditions for growth. The variable identification routine bridges from conceptual understanding to procedural scientific thinking and ensures students approach the observation phase with a testable framework rather than passive watching.	Teacher scans written predictions for scientific reasoning — does the student link warmth and moisture to microbial metabolic needs rather than simply guessing? Teacher listens during pair talk for use of vocabulary from prior lessons (spores, hyphae, moisture, warmth). Misconception to watch for: students may predict that covered containers prevent all growth because they confuse sealing with sterilisation — note this for discussion in the Explain phase.

	tally on the board noting how many students predict warm and uncovered will produce the most growth.	bacteria respond to different conditions.		
Observe Phase  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Exponential Growth and Decay (Flexbook) Source: CK-12  Search ARES for similar readings	Students move into groups of four (six groups maximum). Each group receives: (1) a set of freshly prepared labelled sample containers set up at the start of the day or the previous day by the teacher showing conditions A through F, and (2) two sealed transparent pre-incubated containers Pre-A and Pre-C prepared 2 to 3 days earlier showing visible mould growth on warm uncovered moist ugali and warm uncovered moist bread respectively. Groups also receive a printed or teacher-drawn Observation Data Table with columns for Container Label, Food Type, Conditions, Colour of Growth, Texture Description (fuzzy or slimy or none), Estimated Coverage (none, small patches covering less than 25 percent, medium covering 25 to 75 percent, large covering more than 75 percent), and Smell Description (neutral or sour or no smell). Students spend 8 minutes observing all containers systematically. They use their phones or a class camera to photograph the pre-incubated samples. Students do NOT open any container — all observation is through the transparent walls or sealed clear bags. They complete the data table row by row, describing exactly what they see rather than what they expect. Groups then spend 3 minutes comparing their observations across containers, asking: which condition produced the most	Teacher distributes materials and says: You are now scientists making observations — not guessing, not remembering, looking. Fill in your data table row by row for every container. Do not skip a row even if you see nothing — nothing is data. [WAIT TIME: 8 minutes supervised group observation. Teacher circulates continuously.] While circulating, teacher prompts with: What colour exactly — is it one colour or several? Where does the growth appear first — edge, centre, surface? Does the dry maize biscuit container look different from the ugali container? What does that tell you? Teacher pauses the class halfway and says: Hands up — who can already see a difference between Container A and Container B? [Quick show of hands.] Keep that observation in mind — we will explain it in a moment. After 8 minutes teacher says: Now take 3 minutes — compare across your rows. Which single condition seems to make the biggest difference? Discuss in your group and be ready to give one sentence answer. [WAIT TIME: 3 minutes group discussion.] Teacher listens and notes which groups are identifying moisture versus warmth as the key variable — this is a productive disagreement to surface in the Explain phase.	Systematic structured observation with a pre-formatted data table prevents superficial looking and trains students to distinguish qualitative from semi-quantitative data. Photographing visible mould connects the classroom investigation directly to the phenomenon photographs seen in Lesson 1, making the abstract concept of microbial growth visually and concretely real. The deliberate inclusion of a no-growth result (dry maize biscuit, cool, covered) teaches students that absence of growth is scientifically meaningful data.	Teacher checks that all rows in the data table are completed — blank rows indicate a student skipping inconvenient null results. Teacher listens for precision in descriptive language: fuzzy versus slimy, white versus green versus black. These colour distinctions map back to Lesson 2 where students classified mould types. Listen for whether students spontaneously connect the pre-incubated mould colours to the phenomenon photograph — this indicates deep conceptual linking.

	visible growth and which produced the least?			
Explain Phase  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Exponential Growth and Decay (Flexbook) Source: CK-12  Search ARES for similar readings	Groups share one key observation each in a whole-class gallery-style share-out. Teacher organises responses on the board into a class summary table matching the individual data tables. Once the class table is complete, the teacher guides students through a three-step analysis. Step 1 — Pattern identification: Students compare the moist warm uncovered containers (Pre-A, Pre-C) with the dry cool covered containers and articulate the pattern in one sentence. Step 2 — Cause and effect reasoning: Teacher poses the question: Why does moisture help mould grow? Students draw on their knowledge from Lesson 2 that fungal hyphae absorb nutrients from the substrate through enzymatic digestion, which requires water as a solvent. Step 3 — Phenomenon connection: Teacher displays the phenomenon photograph again and asks students to annotate their data tables with a brief note connecting each condition to the Kisumu ugali scenario. Students write a conclusion paragraph in their notebooks using the frame: The investigation showed that mould grows best when _____. This is because _____. This explains why the ugali and chapati in Kisumu spoiled overnight because _____. A household can prevent this by _____. Finally, the teacher introduces the concept of water activity (without using the technical jargon) by explaining that moulds need free water in the food — this is why dry biscuits resist mould longer than soft ugali, and why salting or	Teacher begins the share-out by saying: Each group give me one observation — the most surprising thing you noticed. No repeats — if another group already said it, tell me something different. [WAIT TIME: allow each group 20 seconds to prepare, then cold-call.] Teacher builds the class table on the board. When the table is complete, teacher says: Look at this table as a whole. In ten seconds, I want you to tell your partner the single biggest pattern you see. Ten seconds — go. [WAIT TIME: 10 seconds.] Teacher takes three responses, then asks: Why does warmth speed up mould growth? Think about what we said in Lesson 2 about how fungi feed. [WAIT TIME: 20 seconds — this is a higher-order link.] After student response, teacher confirms: Exactly — enzymes work faster at warmer temperatures, and the hyphae absorb digested nutrients faster. Now — the big reconnection. [Teacher displays phenomenon photograph.] Was the ugali in Kisumu warm? Was it moist? Was it covered? [WAIT TIME: 10 seconds after each question.] That is your explanation. Write your conclusion paragraph now — 4 minutes silent writing. [WAIT TIME: 4 minutes.] Teacher then says: One more idea. Think about salted dried fish — omena — or dried maize. Why do those last longer without a fridge? [WAIT TIME: 15 seconds.] Teacher introduces the water availability principle simply: Less water available in the food means	The three-step analysis (pattern, cause-and-effect, phenomenon connection) scaffolds students from descriptive observation to mechanistic explanation, mirroring authentic scientific reasoning. The conclusion paragraph frame ensures every student produces a written evidence-based explanation rather than only participating verbally. Connecting the finding to omena and dried maize grounds the abstract variable of water activity in a Kenyan cultural and economic context students encounter daily.	Teacher collects or photographs conclusion paragraphs. The key quality indicator is whether the student connects ALL three elements: experimental condition, biological mechanism and phenomenon explanation. A strong paragraph reads: Mould grows best when conditions are warm, moist and uncovered because fungi need water for enzyme activity and warmth speeds up their metabolism, and spores from the air land and germinate faster. This explains the Kisumu ugali because it was left uncovered and moist in a warm kitchen overnight. A family can prevent this by covering food, refrigerating it or drying it. Students who only describe what they observed without explanation need additional scaffolding in Lesson 8.

	drying meat is a traditional Kenyan preservation technique that reduces available water.	moulds cannot grow. Drying and salting are ancient Kenyan preservation techniques that our ancestors understood through experience — today you understand the science behind them.		
Driving Question Board (DQB) Creation VIDEO: Enzymes Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Exponential Growth and Decay (Flexbook) Source: CK-12 Search ARES for similar readings	In the final 6 minutes of the lesson, students return to the class Driving Question Board which has been building since Lesson 1. Each group receives two sticky notes (or writes directly on the DQB section reserved for Lesson 7). On the first sticky note they write one piece of new evidence their investigation added to answering the driving question: We now know that _____. On the second sticky note they write one new sub-question their investigation raised: We still wonder _____. Possible new evidence notes include: mould grows fastest on warm moist uncovered food; dry substrates resist mould better; covering food slows but may not completely stop mould growth. Possible new sub-questions include: Does refrigeration stop mould completely or just slow it down? Can the same mould that grows on ugali make a person sick? Why are some moulds green and others black — are they different species? The teacher reads several notes aloud and places them on the DQB, explicitly marking which original Lesson 1 predictions have now been confirmed, which have been revised and which remain open. Teacher highlights the progression: in Lesson 1 students predicted the patches came from the air — Lesson 3 confirmed airborne spore transmission —	Teacher distributes sticky notes and says: You have two minutes — two sticky notes per group. First note: one piece of evidence you generated today that helps answer our driving question. Start it with the words We now know. Second note: one question your investigation opened up. Start it with We still wonder. Go. [WAIT TIME: 2 minutes group writing.] Teacher collects notes and reads three or four aloud, placing them on the DQB. Teacher then says: Look at Lesson 1 — what did we predict back then? [Points to original predictions.] Which of those can we now confirm with evidence from our own investigation today? [WAIT TIME: 15 seconds — take two volunteers.] Teacher draws a checkmark next to confirmed predictions and a revision arrow next to those that need updating. Teacher closes the DQB segment by saying: Next lesson we will revise our models and use everything from all seven lessons to answer our driving question completely. Today you made it much easier by generating your own data. Well done.	The sticky-note DQB update ritual maintains the coherence of the eight-lesson storyline by visually marking epistemic progress — students see that their own experimental data has moved the class from wondering to knowing. The distinction between confirmed predictions and revised ones models scientific self-correction and intellectual honesty. Surfacing remaining open questions sustains curiosity and motivates genuine engagement with Lesson 8.	Teacher reads all sticky notes before the next lesson to identify conceptual gaps. High-quality We now know notes will reference specific variables and link to mechanisms. Low-quality notes will be vague (mould needs food). We still wonder notes that show transfer thinking — such as wondering whether refrigeration fully stops growth or whether mould toxins remain after the mould is removed — indicate higher-order readiness for the Lesson 8 model building task.

	and now Lesson 7 shows what conditions allowed those spores to germinate and grow. The driving question banner is visible and teacher points to it: Why did the ugali and chapati grow fuzzy patches overnight? You now have experimental evidence for the overnight part — warm, moist, uncovered, perfect conditions.			
Model Building Phase  VIDEO: Enzymes Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Exponential Growth and Decay (Flexbook) Source: CK-12  Search ARES for similar readings	In the final 3 minutes (brief preparatory phase for the full model revision in Lesson 8), each student opens their Lesson 1 initial model drawing of the phenomenon. They annotate it with a pencil in a different colour by adding two new arrows or labels based on today's investigation: (1) an arrow showing the conditions that allowed spore germination and mould growth on the ugali (label: warm plus moist plus uncovered equals growth); and (2) a crossed-out symbol or note showing what would have prevented growth (label: covered plus cool equals no growth). Students do not redraw the full model — that is reserved for Lesson 8. This quick annotation ensures that the empirical data from today is physically recorded on the model before the lesson ends, so no evidence is lost between lessons. Students who did not draw an initial model in Lesson 1 sketch a simple two-box diagram showing the food surface and the conditions arrow pointing to mould growth.	Teacher says: Open your Lesson 1 model. In a different colour — pen or pencil — add two annotations before you leave today. First, show the conditions that caused growth on the Kisumu ugali. Second, show what would have stopped it. You have 3 minutes. [WAIT TIME: 3 minutes silent individual annotation.] As students work, teacher circulates and says quietly to individuals: Good — can you add the word spore to show where the growth started from? or: What temperature is a Kisumu kitchen at night — think about that as you label warm. Teacher closes the lesson by saying: In Lesson 8 you will redraw your full model from scratch using everything you now know across all seven lessons. You will also write a community message for families in your area. Today you built the evidence. Next lesson you build the explanation.	Annotating the original model rather than replacing it makes student thinking growth visible and honours the iterative nature of scientific modelling. Using a different colour creates a visual record of what was learned through investigation versus what was assumed at the start, reinforcing the epistemic message that science advances through evidence. The brief model annotation also serves as a cognitive consolidation routine that helps transfer new procedural knowledge (variable identification, data patterns) into the conceptual model of the phenomenon.	Teacher collects models at the end of the lesson or photographs them. Quality indicator: does the student annotation correctly identify all three conditions (warmth, moisture, lack of cover) and show a prevention strategy? Does the student use vocabulary from earlier lessons (spores, hyphae, fungi) in the annotation? Models will be returned at the start of Lesson 8 to serve as the starting point for the full model revision.

D. TEACHER REFLECTION

After the lesson, the teacher should consider: (1) Did all students successfully identify at least two conditions that favour mould growth, or were some students only

describing appearance without causal reasoning? If so, begin Lesson 8 with a brief whole-class revisit of the class data table before model revision starts. (2) Were the pre-incubated samples sufficiently visible for meaningful observation? If growth was minimal due to cooler classroom temperatures, prepare samples 3 to 4 days in advance for future iterations and store at 28 to 30 degrees Celsius. (3) Did any students make the connection between the dry maize biscuit result and traditional Kenyan food preservation methods like drying omena or salting nyama choma without prompting? These students are ready for extension tasks in Lesson 8 such as researching fermentation as a preservation technique. (4) Were safety protocols followed — did all students avoid opening containers and wash hands? Note any safety issues for reporting. (5) Review the DQB sticky notes before Lesson 8 to identify the two or three most productive We still wonder questions to open the final lesson, ensuring the narrative arc of the eight-lesson sequence closes in a satisfying and intellectually honest way.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that pre-incubated mould samples on warm moist uncovered ugali and bread showed extensive white, green and black fuzzy growth, while dry cool covered samples showed little or no visible growth. They recorded colour, texture, estimated coverage and smell across six container conditions and noted that the dry maize biscuit showed the least growth under all conditions.
What did I learn?	Students learned that mould growth is favoured by three key conditions: warmth (which speeds up fungal enzyme activity and hyphal growth), moisture (which provides the water needed for enzymatic digestion and nutrient absorption), and exposure to air (which allows airborne spores to settle and germinate on the food surface). Reducing any one of these conditions slows or prevents spoilage, which is why covering food, refrigerating it, or drying it are effective preservation strategies.
How does this explain the phenomenon?	The investigation explained the anchoring phenomenon directly: the ugali and chapati in Kisumu spoiled overnight because the kitchen was warm, the food was moist, and it was left uncovered — all three conditions that maximise mould spore germination and hyphal growth. It also explained traditional Kenyan food preservation practices such as drying omena and salting meat as empirically effective strategies that reduce water availability and temperature, thereby preventing microbial growth without requiring modern refrigeration.

LESSON 8 (40 minutes): Model Building, DQB Synthesis and Community Sensitisation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students consolidate all learning from the microorganisms unit by revising their initial explanatory models, closing the Driving Question Board, and producing a community sensitisation message that applies their scientific understanding to real food safety challenges in Kenyan households.
Knowledge	Students will demonstrate that: (1) fungal moulds, bacteria and viruses differ in structure and mode of transmission; (2) warmth, moisture and air exposure are the key conditions favouring microbial growth on food such as ugali and chapati; (3) covering food, refrigerating it, drying it and maintaining kitchen hygiene are evidence-based prevention strategies; and (4) microorganisms have both harmful and beneficial economic roles in Kenyan communities.
Skills	Students will: revise and annotate a scientific explanatory model of the anchoring phenomenon; synthesise evidence from all eight lessons to close the Driving Question Board; evaluate the accessibility of food preservation strategies for low-income rural and urban Kenyan households; and draft a clear, audience-appropriate community sensitisation message in either English or Kiswahili.
Attitudes	Students will appreciate the value of science in protecting everyday life, demonstrate responsibility toward community health, and show confidence in communicating scientific ideas to non-specialist audiences.
Key Inquiry Question	How are microorganisms important in day-to-day life?
Purpose in Storyline	This final lesson is the culmination of the eight-lesson sequence. Students move from being puzzled observers of fuzzy patches on leftover food to confident scientific explainers who can model the full causal chain — from invisible airborne spores landing on uncovered ugali to infection pathways, prevention strategies and economic impacts — and communicate that understanding to their wider community.
Safety Notes	No hazardous materials are used in this lesson. If pre-incubated mould samples from Lesson 7 are still present in the room, they should remain sealed in their containers throughout the lesson and disposed of safely by the teacher after class by double-bagging before placing in a waste bin. Students should not open containers or handle mouldy food samples.

B. LESSON OVERVIEW

The lesson opens with a brief silent revision of students' individual Lesson 1 prediction drawings, anchoring them back to the original phenomenon of spoiled ugali and chapati in Kisumu. Students then revise their models in groups to reflect everything learned across the unit. Groups present their final models to the class. The class collectively revisits and closes the full Driving Question Board, confirming or correcting all predictions and questions. In the final segment, students draft a short community sensitisation message on food preservation targeted at households in their own locality, fulfilling the KICD project requirement. The lesson closes with a whole-class reflection on the driving question and a brief teacher-led summary.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students are given back their original Lesson 1 prediction	Teacher distributes Lesson 1 prediction sheets. Teacher says:	The predict phase uses a structured prediction-revision	Teacher listens to paired sharing and cold-call responses to gauge

 VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12  Search ARES for similar readings	drawings or written predictions about the fuzzy patches on ugali and chapati. Working individually for 3 minutes, they silently re-read what they wrote in Lesson 1 and annotate it with one phrase: what I thought then versus what I know now. They then share with a shoulder partner. This activates memory of their starting point and creates cognitive contrast with their current understanding, setting up meaningful model revision.	Before we build our final models, I want you to look at what you thought in Lesson 1. Read your original prediction silently. Then write just one phrase at the top of the page: one thing you thought then that you now know was incomplete or wrong. You have 2 minutes to do this quietly. [WAIT TIME: 2 minutes of silent individual work. Teacher circulates and reads over shoulders without commenting.] After 2 minutes, teacher says: Share your one phrase with your partner. You have 1 minute each. [WAIT TIME: 2 minutes of paired sharing.] Teacher then cold-calls three students and records their contrasting phrases on the board under two columns labelled LESSON 1 THINKING and LESSON 8 THINKING. Teacher says: These changes in your thinking are exactly what science does. Today we make those changes visible in a final model.	protocol. By returning to original predictions, students engage in metacognitive reflection — a core scientific practice. The two-column teacher recording makes the conceptual progression explicit and public, honouring prior thinking while demonstrating growth.	whether students can articulate specific conceptual changes. If students say only vague phrases such as I know more now, teacher probes by asking: Can you name one specific thing that changed? This reveals whether the unit has produced durable conceptual understanding or surface familiarity.
Observe Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of four receive a Model Revision Sheet — a blank A3 template divided into six labelled boxes: (1) What the fuzzy patches are and their types, (2) Where the microorganisms came from and how they arrived on the food, (3) Conditions that allowed them to grow overnight, (4) How microorganisms can enter the human body and cause infection, (5) How we can prevent spoilage and infection in a Kenyan household, (6) One beneficial role of microorganisms relevant to Kenya. Students spend 10 minutes revising and completing their model collaboratively, drawing annotated diagrams and writing brief captions. They must use	Teacher distributes Model Revision Sheets and says: Your group has 10 minutes to build your best final model of the phenomenon. Every box must be filled. You must use evidence from our investigations and research — not just opinion. Write the lesson number in brackets next to any fact you include so I can see where your evidence comes from. [WAIT TIME: 10 minutes. Teacher circulates continuously, spending no more than 2 minutes with any one group. Teacher uses targeted prompts rather than giving answers.] Suggested teacher prompts during circulation: To a group that has left box 3 empty — What did your investigation in	The six-box model template scaffolds systems thinking by requiring students to connect microorganism identity, transmission, growth conditions, infection pathways, prevention and beneficial roles into a single coherent explanatory framework. The instruction to cite lesson numbers operationalises evidence-based reasoning. The Kisumu kitchen framing continuously anchors abstract science to the concrete phenomenon.	Teacher photographs two or three group models mid-activity using a phone or tablet for later whole-class display. During circulation, teacher notes which boxes are consistently left incomplete or contain misconceptions — for example, students who conflate bacteria and viruses or who list prevention strategies without explaining their scientific basis. These gaps are addressed during the Explain Phase.

	evidence from at least three different lessons in their unit (e.g., Lesson 2 types, Lesson 3 transmission, Lesson 7 investigation findings).	Lesson 7 tell you about what conditions mould needs? To a group that has only written text and no diagrams — Can you draw the spore dispersal process so someone who has never studied this could understand it? To a group that has not connected back to the phenomenon — Your model should explain the ugali and chapati in that Kisumu kitchen. Does your model do that? At 8 minutes, teacher gives a 2-minute warning: Two more minutes — check that the family in the Kisumu kitchen could use your model to understand what happened and what to do next.		
Explain Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12  Search ARES for similar readings	Three groups present their final models to the class — each group has a maximum of 2 minutes to walk the class through their six boxes using their A3 sheet held up or taped to the board. The presenting group points to each box in sequence and gives one sentence per box. After all three presentations, the class participates in a structured gallery critique: each group writes one strength and one question on a sticky note and passes it to the presenting group. Teacher then leads a 4-minute whole-class consolidation discussion, drawing out the consensus scientific explanation of the phenomenon and correcting any remaining misconceptions. The class then moves to the community sensitisation task: each group drafts a 3-sentence message (in English or Kiswahili) addressed to a specific audience — either a market vendor in Kisumu, a parent in a rural Nyeri household, or a school tuck-shop operator —	Teacher introduces presentations by saying: Each group has 2 minutes. Walk us through your model box by box. Speak to the class, not to the paper. I will time you. [WAIT TIME: 2 minutes per group, teacher keeps time firmly.] After each presentation, teacher says: Write your sticky note now — one strength, one question — and hand it to that group. [WAIT TIME: 30 seconds for sticky notes after each presentation.] After all three presentations, teacher says: Looking at all three models, what is the one explanation that ALL groups agree on about why the fuzzy patches appeared on the ugali overnight? [WAIT TIME: 10 seconds of silence before taking responses.] Teacher records the consensus explanation in one sentence on the board — for example: Airborne fungal spores landed on uncovered warm moist ugali and germinated overnight to form visible mould colonies. Teacher then addresses any misconception observed during	The structured 2-minute presentation format trains scientific communication under constraints. The sticky-note critique protocol builds peer evaluation skills and distributes feedback beyond the teacher. The consensus sentence strategy — where the class co-constructs one agreed explanation — is a key sensemaking move that distinguishes scientific consensus from individual opinion. The community sensitisation task applies the science to authentic Kenyan contexts, making the knowledge functional and transferable.	Teacher listens for accuracy and causal language during presentations — students should use terms such as spore dispersal, germination, conditions for growth, and prevention rather than vague language. The sticky-note questions reveal remaining confusions across the class. The sensitisation messages are collected and scanned by the teacher after class to assess whether students can translate technical knowledge into practical, audience-appropriate advice — a key CBE competency.

	advising them on how to prevent food spoilage and protect their family or customers from microbial infection.	model circulation — for example: I noticed some groups wrote that the mould came from inside the ugali. What does our Lesson 7 investigation evidence tell us about that? [WAIT TIME: 5 seconds.] For the sensitisation message, teacher says: You are now scientists writing for your community. Choose one of these three audiences. Write three sentences that are scientifically accurate, practical for that person, and easy to understand. You have 4 minutes. [WAIT TIME: 4 minutes of group writing.]		
Driving Question Board (DQB) Creation  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12  Search ARES for similar readings	The full Driving Question Board from Lesson 1 — with all accumulated questions, predictions and updates from Lessons 1 through 7 — is displayed at the front of the class. The teacher reads out each original question and prediction in sequence. For each one, the class votes using a three-sign system: thumbs up for confirmed, thumbs sideways for partially answered or revised, and thumbs down for disproved. A student volunteer places a green, orange or red sticky dot on each DQB card accordingly. Any question that remains unanswered after class voting is discussed briefly — the class decides whether it is beyond the scope of this unit or whether a partial answer can be offered based on available evidence. The board is then photographed for the school science record. Students spend 1 minute individually writing in their notebooks: The driving question asked why the ugali and chapati grew fuzzy patches. My full answer now is...	Teacher gestures to the DQB and says: This board has been our scientific journey since Lesson 1. Every question, every prediction, every update is here. We are going to close it together now. I will read each card. You vote with your hands. A volunteer places a dot. We move on. [Teacher reads cards briskly, pausing 5 seconds for each vote.] [WAIT TIME: 5 seconds per card for class consensus.] If a card receives mixed responses, teacher asks: Who thinks this is only partially answered? What part is still unclear? [WAIT TIME: 5 seconds.] After the full board is dotted, teacher says: Look at how many green dots there are. You arrived in Lesson 1 not knowing what those fuzzy patches were. Look at what you can explain now. Now write your full answer to the driving question in your notebook. One minute. [WAIT TIME: 60 seconds of silent individual writing.] Teacher then cold-calls two students to read their answers aloud, affirming scientific accuracy and precise language.	The three-dot voting system makes collective knowledge visible and democratic — students see that science involves revising predictions, not just confirming them. The presence of orange and red dots is treated as a sign of scientific integrity, not failure. The individual notebook response to the driving question is a high-stakes metacognitive synthesis task that requires students to consolidate the entire unit into one coherent causal explanation.	The distribution of green, orange and red dots gives the teacher a rapid whole-class diagnostic of which concepts have been mastered and which remain uncertain. The individual notebook responses are the primary summative evidence of each student having answered the driving question. Teacher collects notebooks or photographs responses for record-keeping and parent-reporting purposes.

Model Building Phase  VIDEO: Populations, communities, and ecosystems Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Fungi (Flexbook) Source: CK-12  Search ARES for similar readings	In the final 5 minutes, groups share their community sensitisation messages aloud — one sentence each, rotating around the group so every member speaks. The class applauds each message. Teacher then leads a closing reflection using three questions written on the board: (1) What is the most important thing you will do differently at home because of what you learned in this unit? (2) What is one question this unit raised that you want to investigate further? (3) How does understanding microorganisms connect to being a responsible citizen in Kenya? Students respond in a rapid round-robin — teacher points to students and they give one-sentence answers without prior preparation. Teacher closes with a brief contextualising statement connecting the learning to broader Kenyan public health challenges.	Teacher says: Before we close, each group shares one sentence from your community message. One sentence, one person, then pass to the next group member. Let us hear every voice. [WAIT TIME: teacher allows each speaker to finish before pointing to the next.] After messages are shared, teacher writes the three reflection questions on the board and says: I am going to point to you at random. One sentence. No preparation. What is the most important thing you will do differently at home? [WAIT TIME: 3 seconds per student before pointing to the next.] After the round-robin, teacher closes by saying: In 2023, cholera outbreaks linked to contaminated water and uncovered food affected communities around Lake Victoria. In 2025, food vendors in Kisumu still lose income daily to mould spoilage. The invisible world of microorganisms is not abstract science — it is in your kitchen, your market, your community. You now have the knowledge to protect your family and educate others. That is what science is for. Well done.	The round-robin reflection protocol ensures every student articulates a personal and civic connection to the science, moving beyond knowledge recall to values and agency. The teacher closing statement grounds the unit in named, real, recent Kenyan public health contexts — cholera near Lake Victoria, food vendor spoilage losses in Kisumu — reinforcing that science education in Kenya is purposeful and locally relevant. The community sensitisation message task fulfils the KICD requirement for students to apply learning in authentic community contexts.	The round-robin responses allow the teacher to assess three competencies simultaneously: personal application of knowledge (question 1), scientific curiosity and next-step thinking (question 2), and citizenship orientation (question 3). Students who struggle with question 1 may need follow-up support in connecting abstract science to daily behaviour. The sensitisation messages are retained as portfolio evidence of the CBE project component for this sub-strand.
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D. TEACHER REFLECTION

After the lesson, the teacher should reflect on the following questions: Did the revised models show genuine conceptual growth from Lesson 1, or did groups reproduce surface vocabulary without deep causal understanding? Were students able to connect their Lesson 7 investigation findings — warmth, moisture, air exposure — into their final models as evidence rather than as isolated facts? Did the community sensitisation messages demonstrate audience awareness, or did students default to generic scientific language that would not be understood by a market vendor or rural parent? Which DQB cards received the most orange or red dots, and what does that reveal about which concepts in the unit need reinforcement in future cycles? Were all students able to articulate a personal behaviour change connected to the science, or did the civic connection feel performative for some learners? Finally, the teacher should note whether the 40-minute duration was sufficient — if model revision ran long, the sensitisation message drafting may have been rushed, and a 5-minute extension or a split across two sessions should be considered for future delivery.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed their own revised model sheets alongside pre-incubated mould samples from Lesson 7, the full Driving Question Board with all accumulated questions and predictions from eight lessons, and three group model presentations — all anchored to the original photographs of spoiled ugali and chapati from the Kisumu kitchen.
What did I learn?	Students confirmed that the fuzzy patches on the ugali and chapati were fungal mould colonies that grew from airborne spores which landed on uncovered, warm and moist food overnight; that bacteria and viruses also cause disease through specific transmission routes; that covering food, refrigerating it or drying it prevents spoilage by removing conditions microorganisms need to grow; and that microorganisms also play beneficial roles in Kenyan food production, agriculture and medicine.
How does this explain the phenomenon?	Students explained the full causal chain of the phenomenon — from invisible airborne fungal spores dispersing through the kitchen air, settling on uncovered ugali and chapati, germinating in warm moist conditions overnight, forming visible white, green and black mould colonies — and communicated evidence-based food preservation advice to specific community audiences in Kenya, demonstrating that science understanding can protect health and economic wellbeing at the household level.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.